

Game Design Document

GameName: Reversal Flip

TeamName: Flip

Team Members

Name	Roles	E-mail & GitHub ID
<i>Xiaoyu Ke</i>	<i>Team Captain, Level Design</i>	<i>xiaoyuke@usc.edu/ kyleke599@gmail.com KyleKe599</i>
<i>Sherry Wang</i>	<i>User Interface, Product Manager</i>	<i>wangxuen@usc.edu sherry.wang1579@gmail.com sherrywang615</i>
<i>Xinlu Xu</i>	<i>Product Manager, Physics</i>	<i>xinluxu@usc.edu sabinaxu1028</i>
<i>Yang Li</i>	<i>Product Manager, Level Design, Physics</i>	<i>lyang835@usc.edu leahyangl</i>
<i>Jiayi Zhang</i>	<i>Stenographer, Physics (Magnetic Effect)</i>	<i>jiayiz99@usc.edu/ lagrange1512@gmail.com LaGrange151235</i>
<i>Aryan Avinash Sawant</i>	<i>Stenographer, Level Design</i>	<i>aryanavi@usc.edu aryanavi6802</i>

Links

Unity Project GitHub Repository	https://github.com/CSCI-526/main-flip
Playable Alpha Progress Check Build	https://lagrange151235.github.io/CSCI526/FinalProject/flip_v.0.4/
Playable Alpha Build	https://lagrange151235.github.io/CSCI526/FinalProject/flip_v.0.5/
Alpha Gameplay Video	https://youtu.be/E01Oq_gMjYo
Alpha Survey	https://docs.google.com/forms/d/1tFdNkrK28RhN6apKxOBA-9381a8dfSnaSsRTlRluDKo/viewform?edit_requested=true
Playable Beta Progress Check Build	https://lagrange151235.github.io/CSCI526/FinalProject/flip_v.0.7.7/
Playable Beta Build	https://lagrange151235.github.io/CSCI526/FinalProject/flip_v.0.8.6/
Beta Gameplay Video	https://youtu.be/ksNkb1eAYEA
Beta Presentation Slide	https://docs.google.com/presentation/d/1jw3AbZy8XInDtsjburkD9wEUAV1lrY9ChL4AuclZSLQ/edit?usp=sharing
Beta Survey	https://docs.google.com/forms/d/e/1FAIpQLSf8JsjJ2TretBPIrVFbUyHv5RZDHgLVBs4NQx7795wDPyBjRQ/viewform
Playable Gold Progress Check Build	https://lagrange151235.github.io/CSCI526/FinalProject/flip_v.0.9.1/
Playable Gold Build without Art	https://lagrange151235.github.io/CSCI526/FinalProject/flip_v.0.10.4_without_art/
Playable Gold Build with Art	https://lagrange151235.github.io/CSCI526/FinalProject/flip_v.0.10.4_with_art/
Gold Gameplay Video	https://youtu.be/v0fd2-XUp_U
Gold Presentation Slide	https://docs.google.com/presentation/d/1qt1A7kcanVQk84KfQpHDdn1TDC6LIBSeSuXa8jansX0/edit?usp=sharing

*Note: It is required to have **different build links** for each of these sections. **Do NOT overwrite** a previous build with a new one as it will cause issues during grading.*

Game Introduction

Logline

A 2D-platformer where players use simple controls to switch magnetic polarity and manipulate anti-gravity to escape a maze of magnets.

Player Goals / Win Conditions

By using magnetism and gravity to solve maze and complete movement.

Game Description

Our game aims to create a new gameplay experience by replacing traditional player controls with simple mechanics. It is a 2D platformer that removes conventional movement and jumping, substituting them with two core systems — magnetism and gravity. This combination completely redefines how players navigate and interact with the environment, making it the key twist of our game.

In the current prototype, players move and solve puzzles solely by manipulating magnetic polarity and gravity direction. They can attract or repel from magnetic surfaces, or switch gravity to move vertically across the level. Unlike traditional platformers, where progress depends on precise jumping and running, our game focuses on mastering these two forces to traverse and trigger mechanisms, offering a refreshing and intuitive platforming experience.

Controls

For Laptop/Computer:

- *Space: Gravity Control*
- *Shift: Magnetism Control*

For Mobile Device:

- *Gravity Switch Button: Gravity Control*
- *Polarity Switch Button: Magnetism Control*

Detailed Design

Players control a character that switches magnetic polarity and gravity to move through a maze. By using magnetism and gravity to traverse obstacles and activate mechanisms, they must find a path from the start to the end of the maze without traditional movement or jumping.

Game Mechanics (How to Play)

Movement mechanic: Use the keyboard to control the character's magnetism and gravity.

Goal mechanic: Navigate through the maze and reach the win gate to complete the level.

Fail mechanic: The player loses if they touch spikes or fall out of the game area.

Game Mechanic Research

<Game mechanic 1: Grabbing> Sherry Wang

Game Title	<i>Fall Guys</i>
Mechanic/System	<i>Grabbing mechanic - hold objects or other players</i>
Interest	<i>This mechanic allows players to physically interact with others in the game world. They can grab and stop opponents from moving forward or hold onto platforms to avoid falling off. It adds a new layer of strategy and tension, elevating Fall Guys beyond a traditional 3D platformer by introducing direct player-to-player interaction and tactical control.</i>
Interaction with other mechanics	<i>The grabbing mechanic interacts with player movement, physics, and environmental obstacles. Timing and positioning are important when jumping between platforms or grabbing objects. In some mini-games, players can grab bombs and throw them at opponents to knock them off the arena.</i>
Innovation	<i>Players could grab onto another player's feet when hanging on the edge of a platform. By swinging while holding on, they could cause the opponent above them to lose balance and fall. This creates new offensive and defensive strategies. Players could use it to pull others down or to save themselves by holding onto someone else. In team modes, it could even enable cooperative climbing, where players help each other reach higher platforms.</i>

<Game mechanic 2: Color-Switching> Xiaoyu Ke

Game Title	<i>Chameleon Run</i>
Mechanic/System	<i>Color-switching mechanic</i>
Interest	<i>The color-switching mechanic in this game requires players to make instant decisions, creating both tension and flow. It also keeps the controls simple, relying on a single mechanic that defines the entire gameplay experience.</i>

Interaction with other mechanics	<i>The color-switching mechanic interacts closely with movement and jumping physics. Players must time their jumps while simultaneously switching colors in mid-air.</i>
Innovation	<i>Chameleon Run simplifies the traditional 2D platformer by focusing only on color switching and jumping, completely removing the usual directional movement. Similarly, in our game, players only need to manage gravity and magnetism without worrying about other actions. Both color switching and polarity switching share the same design philosophy — maintaining simple controls while allowing players to experience the game in a more intuitive and engaging way.</i>

<Game mechanic 3: Trigger> Yang Li

Game Title	<i>Level Devil</i>
Mechanic/System	<i>Trigger mechanic</i>
Interest	<i>The triggers help guide the player toward reaching the end gate, while adding fun and challenge. The player needs to think about their usage and use them correctly.</i>
Interaction with other mechanics	<i>This gameplay mechanic interacts with player movement and environment behavior. Walking or jumping into certain zones or buttons may trigger hidden traps, moving platforms or invisible hazards, which creates unpredictable and surprising level change.</i>
Innovation	<i>In the Level Devil, triggers are activated by left, right and jump movement, which is the typical mechanic in 2D platformers. However, since all player's movements are replaced by gravity and magnet polar manipulation in our game, the triggers can be triggered with the combination of gravity or magnet polar change.</i>

<Game mechanic 4: Anti-gravity> Xinlu Xu

Game Title	<i>VVVVVV</i>
Mechanic/System	<i>Anti-gravity</i>
Interest	<i>The gravity inversion mechanic helps change how players think about movement and level navigation. Instead of jumping, the player flips gravity to walk on the ceiling.</i>
Interaction with other mechanics	<i>Gravity inversion shapes both movement and puzzle design. Each flip turns floors into ceilings, forcing players to plan timing</i>

	<i>and position carefully. Without a jump button, movement depends entirely on precise flips, momentum control, and reading hazards, turning simple navigation into a challenging rhythmic puzzle.</i>
Innovation	<i>I could twist this mechanic by placing it in a puzzle game or combat game instead of a platformer. For example, gravity flipping could affect projectiles, enemies, or fluids in the environment, which allows the player to fight or solve problems by changing gravity for the whole scene rather than just their own movement.</i>

<Game mechanic 5: Time reverse> Jiayi Zhang

Game Title	<i>Braid</i>
Mechanic/System	<i>Time reverse</i>
Interest	<i>Braid gives the player the power to reverse time and rewind actions, even after death. This time reversal ability makes otherwise impossible puzzles feasible.</i>
Interaction with other mechanics	<i>Braid builds upon the fundamental controls of most platformers – running, jumping, and climbing – to navigate its levels. Player can defeat enemies by jumping and stomping on them.</i> <i>Additionally, there are different time-based mechanics in different levels (worlds):</i> <i>- World 2: Introduces the basic time-reversal ability.</i> <i>- World 3: Player still has time -reversal ability, but objects surrounded by a green glow are unaffected by time manipulation.</i> <i>- World 4: The flow of time is tied to the player's horizontal movement. Player moves right, time flows forward. Player moves left, time reverses.</i> <i>- World 5: A shadow of the player appears after rewinding time. The shadow performs the actions that the player rewound. Objects highlighted in violet can interact with both player and his shadow.</i> <i>- World 6: Player gets a magic ring that warps time around it when dropped. The closer any moving object (including the player) is to the ring, the slower time passes for it.</i> <i>- World 1 (Final world): Time flows in reverse. Rewinding time returns the flow of time to its normal direction.</i>
Innovation	<i>Braid focuses more on platforming than on combat. By combining time reversal mechanic with more combat gameplay, we can create a fun Metroidvania game.</i>

	1. In this game, the player is not permitted to rewind time at will. The ability is activated only upon death. 2. Upon death, the player is reborn at the nearest checkpoint. Concurrently, some temporal shadows of the player appear, which automatically replay the player's previous actions starting from some specific set points defined by the rewind sequence. 3. Player can create multiple shadows after each death. However, each shadow of the player consumes one slot of energy, thereby limiting the player to a finite number of shadows at any given time. With these shadows replicating past actions, the player can coordinate complex strategies to defeat powerful enemies and bosses that would otherwise be insurmountable alone.
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<Game mechanic 6: Dash> Aryan Avinash Sawant

Game Title	Celeste
Mechanic/System	Dash
Interest	The Dash is an instantaneous, omni-directional burst of movement. It is a limited resource, restricted to a single use in the air before being refreshed by touching solid ground or a unique environmental object (e.g., a Blue Crystal). The system transforms basic movement into a tactical resource management challenge. The fixed constraint (one dash) forces the player to approach platforming obstacles not just with timing, but with deliberate, sequential planning, creating depth from a single core action.
Interaction with other mechanics	The dash interacts with nearly every other mechanic in the game: • <i>Wall Climb (Stamina):</i> The dash can be used to quickly traverse a wall, allowing the player to conserve their limited wall-climbing stamina. • <i>Momentum/Physics:</i> A dash can carry momentum into a jump (or a jump can carry momentum into a dash), allowing for greater total distance or height than either move could achieve alone (e.g., a "wave-dash" maneuver). • <i>Environment (Crystals/Orbs):</i> Certain environmental objects (blue crystals, dream blocks) are specifically designed to interact with the dash, often refreshing it mid-air and allowing for highly complex, multi-dash platforming sequences.

Innovation	<i>Dash Refills via Combat: The primary way to regain a Dash charge is not by touching 'ground,' but by successfully parrying an enemy attack or dealing some damage to the enemy. This creates a reward for defensive as well as attacking maneuvers.</i> <i>The Momentum Attack: If a unit uses a Dash to move and then immediately ends its movement next to an enemy, the next standard attack is granted bonus damage (a crit or armor-pierce), consuming the remaining Dash charge for the turn.</i>
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Matrices

Twist & Mechanics Matrix

Mechanics	Description	Interaction with Twist	Affected Genre Elements	Supports
<i>Force Field Manipulation (gravity and player's magnetic polarity)</i>	<i>The player can manipulate the direction of gravity and the character's magnetic polarity.</i>	<i>Manipulating of gravity and magnet polar takes the place of the traditional movement mechanic.</i>	<i>Player movement and jumping</i>	-
<i>Moving & Rotating Magnetic Platforms</i>	<i>Platforms that move according to the direction of gravity, or rotate when a trigger is activated.</i>	<i>The player has to change its polarity in order to stick to the platform.</i>	<i>Movement</i>	<i>Force Field Manipulation</i>
<i>Trigger</i>	<i>A switch, when activated, causes corresponding changes in platforms, such as rotation or movement, or triggers other changes.</i>	<i>Accessible via creative manipulation of gravity and magnet polar.</i>	<i>Platform movement</i>	<i>Force Field Manipulation , Moving & Rotating Magnetic Platforms</i>
<i>Traps/ Hazard</i>	<i>Elements like spikes that instantly defeat or damage the player upon contact, creating an obstacle to traversal.</i>	<i>Hazards like spikes are fixed elements that must be avoided via precise Force Field Manipulation (gravity and</i>	<i>Player Death and Platforming Challenge</i>	<i>Force Field Manipulation , Moving & Rotating Magnetic Platforms, Trigger</i>

		<i>polarity shifts) and timing with Moving Platforms.</i>		
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Mechanic Interaction Matrix

Subject / Object	Force Field Manipulation	Moving & Rotating Magnetic Platforms	Trigger	Traps/ Hazard
Force Field Manipulation	-			
Moving & Rotating Magnetic Platforms	Some platforms react to the change in gravity's direction and fall accordingly	-		
Trigger	Player explores the levels and activates triggers by manipulating his/her own magnetic polarity and the direction of gravity	Player must touch the corresponding trigger to activate the moving/rotating platform	-	
Traps/ Hazard	Traps are unaffected by the player's polarity or gravity; the player's Force Field Manipulation does not influence them.	The movement/rotation of a platform may push the player directly into a trap, creating an obstacle with a precise time window.	There are traps near the triggers to make it difficult for the player to collect the trigger.	-

Single Mechanic Challenge Matrix: Force Field Manipulation

Challenge	Location	Difficulty	Player Must Learn	Instructions	Required Skills
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Challenge 1	Tutorial	Easy	How to perform basic polarity shift (stick) and gravity flip (change direction).	Display on-screen prompts for Shift (polarity) and Space (gravity).	Basic movement controls and button press.
Challenge 2	Middle of the first level	Easy	Using polarity shifts to rapidly traverse two adjacent magnetic surfaces (Polarity Spam).	Provide a hint detailing the five-press solution for the Sticky Keys issue and the benefits of continuous polarity shift.	Basic movement and precise timing of the Shift key.
Challenge 3	Beginning of second level	Medium	Combining a gravity flip mid-jump to extend distance, followed by a quick polarity shift to stick.	Contextual hints demonstrating the velocity preservation during a mid-air gravity flip.	Precision in movement, timing, and velocity awareness.
Challenge 4	Middle of second level	Medium	Alternating polarity rapidly while climbing a vertical shaft against a time limit or moving obstacle.	Encourage experimentation with rapid input sequences and maintaining rhythm.	Coordinated movement and spatial awareness.
Challenge 5	Third Level	Hard	Navigating intricate platforming that requires chained gravity/polarity flips in succession without touching neutral ground.	Minimal guidance, player-driven exploration of advanced movement sequences.	Advanced movement techniques and strategy.

Single Mechanic Challenge Matrix: Moving & Rotating Magnetic Platforms

Challenge	Location	Difficulty	Player Must Learn	Instructions	Required Skills
Challenge 1	Middle of the first level	Tutorial	How to use polarity to stick to a platform and ride it safely across a gap.	Display on-screen prompt: "Use Polarity to ride."	Basic stick and movement timing.
Challenge 2	End of the first level	Easy	Timing a jump off a moving platform to land on a static target.	Provide hints on judging speed and jump trajectory.	Basic timing and spatial judgment.
Challenge 3	Beginning of second level	Medium	Dealing with platforms that rotate or move in a complex (non-linear) pattern while maintaining polarity.	Contextual hints showing the rotational path of the platform.	Precision in movement and sustained button input (Shift key).
Challenge 4	Middle of second level	Medium	Using a gravity flip to make a magnetic platform fall into a necessary position.	Encourage experimentation: "How can you use gravity to change the environment?"	Coordinated movement, spatial awareness, and environmental manipulation.
Challenge 5	Third Level	Hard	Chaining movements across multiple rotating and moving platforms, each requiring a different polarity or gravity state.	Minimal guidance, player-driven exploration of movement sequence planning.	Advanced movement techniques and complex timing strategy.

Single Mechanic Challenge Matrix: Trigger

Challenge	Location	Difficulty	Player Must Learn	Instructions	Required Skills
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Challenge 1	First Level	Easy	The basic function: stepping on a Trigger causes a corresponding platform to move or appear.	Display on-screen prompt showing the effect (platform movement) when the Trigger is activated.	Basic platform interaction.
Challenge 2	Middle of the second level	Easy	Activating a Trigger that requires a simple Force Field Manipulation to reach.	Provide hints on using polarity to walk on a wall or ceiling to access the Trigger.	Basic Force Field Manipulation and navigation.
Challenge 3	End of the second level	Medium	Activating a Trigger that has a non-local or delayed effect.	Contextual hints drawing attention to a distant platform or a time delay bar.	Precision in movement and observing distant environmental changes.
Challenge 4	Beginning of the third level	Hard	Activating multiple Triggers in a rapid or specific sequence to solve a puzzle.	Encourage experimentation with which Trigger affects which element and in what order.	Coordinated movement and sequential logic/problem-solving.
Challenge 5	Third Level	Hard	Activating a Trigger that also activates a new Traps/Hazard , requiring immediate platforming execution.	Minimal guidance, challenging the player's quick reaction time.	Advanced movement techniques and high-speed decision-making.

Single Mechanic Challenge Matrix: Traps/Hazards

Challenge	Location	Difficulty	Player Must Learn	Instructions	Required Skills
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Challenge 1	Middle of the first level	Easy	Recognizing hazards (spikes) and using a single Gravity Flip to bypass them.	Display on-screen prompt: "Avoid hazards by manipulating gravity."	Basic recognition and Gravity Flip execution.
Challenge 2	End of the first level	Easy	Navigating a series of static spikes requiring precise Polarity Shifts along a magnetic wall.	Provide hints on maintaining a rhythm with the Polarity Shift.	Basic timing and precision in movement.
Challenge 3	Beginning of second level	Medium	Dodging hazards while being moved by a Moving Magnetic Platform .	Contextual hints: "Watch the platform's path relative to the hazards."	Precision in movement and spatial awareness.
Challenge 4	End of second level	Hard	Bypassing a complex maze of hazards that requires a sequence of Polarity and Gravity shifts.	Encourage experimentation to find the most efficient path through the trap configuration.	Coordinated movement, spatial awareness, and pathfinding.
Challenge 5	Third Level	Very hard	Bypassing a complex maze of hazards that requires a sequence of Polarity and Gravity shifts.	No guidance, challenging the player to find the connection between the Trigger and the Hazard's state.	Advanced movement and high-speed execution under pressure.

Tutorialization implementation for Mechanics (Beta):

1. Tutorial Level: Isolation Phase

- **Part 1: Gravity Control**
 - **Goal:** The player must learn how the y-axis flip affects their movement and momentum.
 - **Design:** Simple static platforms. The player must simply flip gravity to traverse pits or reach high ledges. The segment should be forgiving, with no instant-death hazards.
- **Part 2: Magnetic Polarity Control**

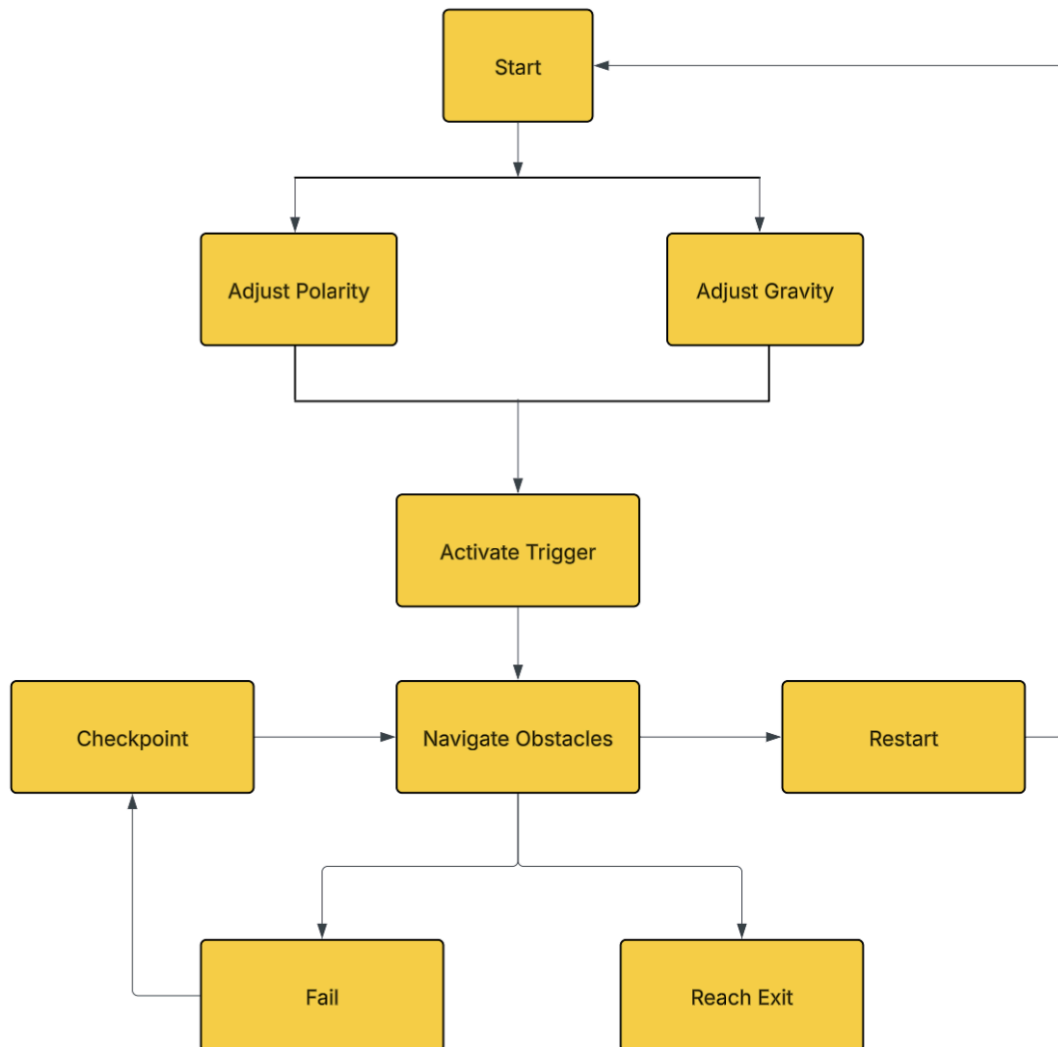
- **Goal:** The player must learn the concept of attraction and repulsion and how to change their own polarity state.
- **Design:** Introduce a basic magnetic track or block that requires the player to switch polarity to either cling to it (attraction) or push off it (repulsion) to cross a gap.

2. Level 1: Combination & Consequence Phase

- **Part 1: Basic Combination & Triggers**
 - **Goal:** Use both Gravity and Polarity controls in sequence, and introduce the concept of environmental interaction via Triggers.
 - **New Elements:**
 - **Moving & Rotating Magnetic Platforms:** Platforms that require a polarity change to ride/climb *while* potentially requiring a gravity flip to exit.
 - **Triggers:** A simple button that, when activated, causes a nearby platform to move or rotate a short, predictable distance/angle.
- **Part 2: Hazard Introduction (Spikes)**
 - **Goal:** Introduce the first instance of lethal failure and require precision evasion using the combined mechanics.
 - **Design:** Areas where spikes are easily avoided using a slow, deliberate gravity flip or a timely magnetic repulsion/attraction, training the player to respect the hazard.

Sketches and Diagrams

Gameplay Loop



Mechanic/Level Sketches

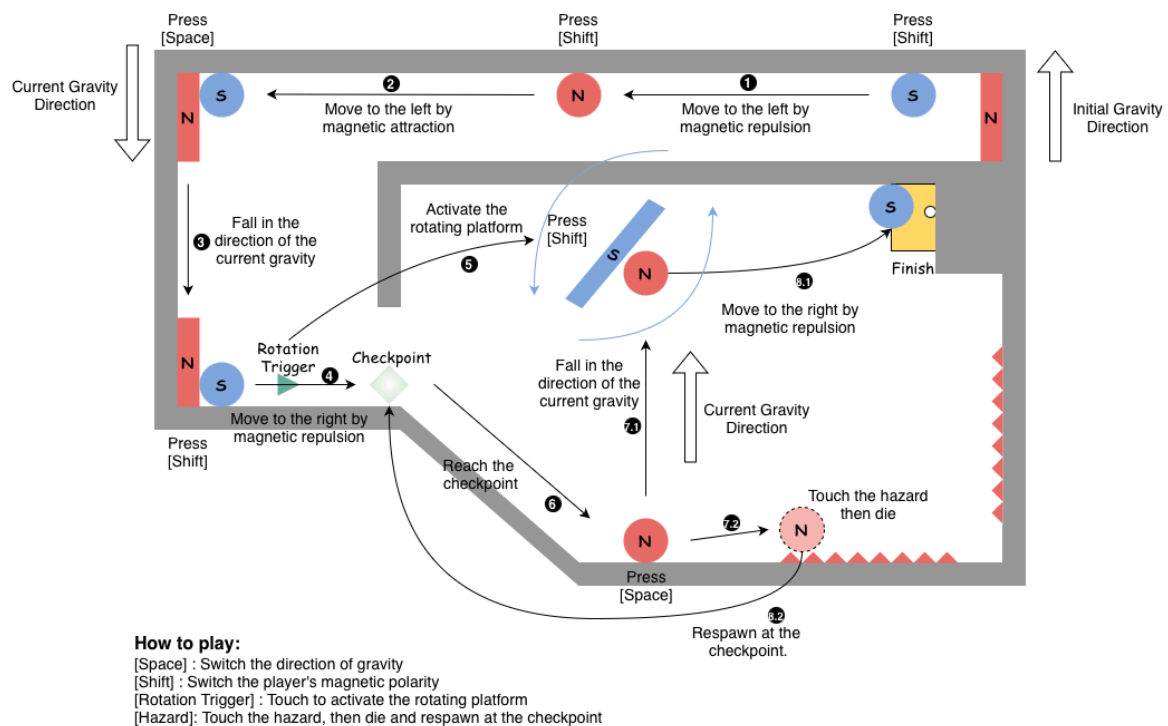


Figure: Mechanic Sketch

This sketch illustrates how players control movement by manipulating gravity direction with *Space* and switching magnetic polarity with **Shift**, enabling attraction or repulsion from magnets. Trigger objects shown in the sketch activate mechanical changes such as rotating magnetic platforms, opening new traversal routes. Hazards cause player death and respawn at the most recent checkpoint.

Level Designs (Beta)

- Gravity Tutorial

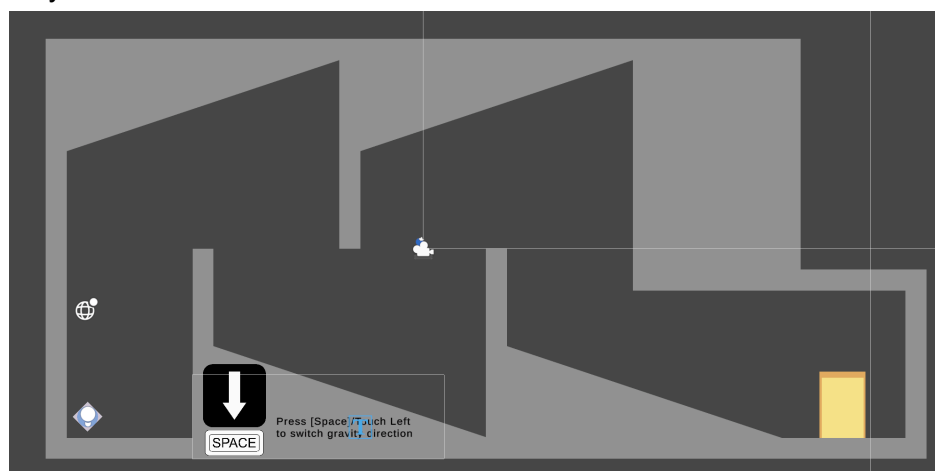


Figure: Gravity Tutorial

The Gravity Tutorial teaches players how to change the direction of gravity using **Space**, introduces the use of the recharger, and explains the basic goal of the game.

Through repeated gravity switching, players quickly develop familiarity with manipulation of gravity.

- Magnetic Tutorial

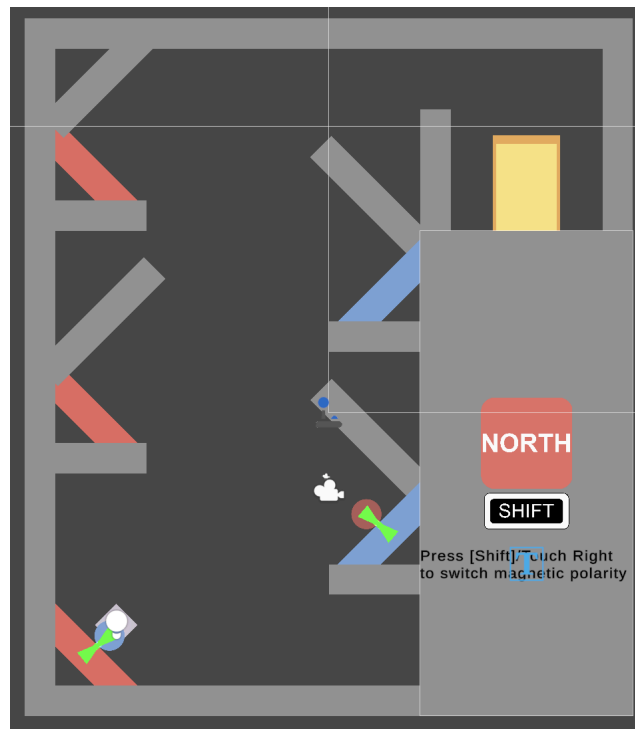


Figure: Magnetic Tutorial

The Magnetic Tutorial teaches players how to switch player's magnetic polarity using **Shift**. Through repeated polarity switching in a safe environment, players quickly develop familiarity with how magnetic attraction and repulsion work.

- Level 1
Sketch:

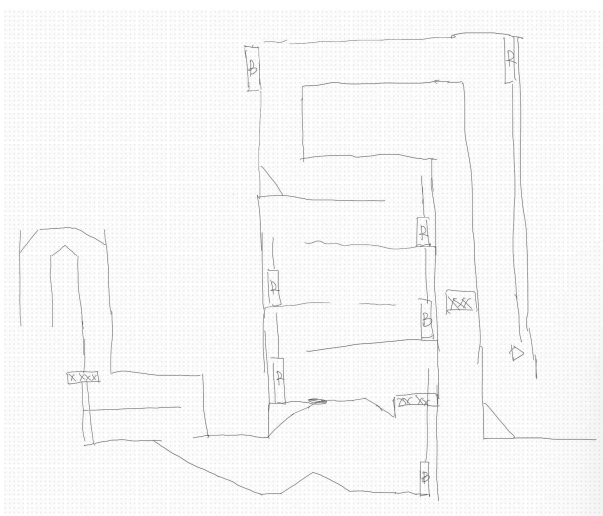


Figure: Level 1 Part 1 Sketch

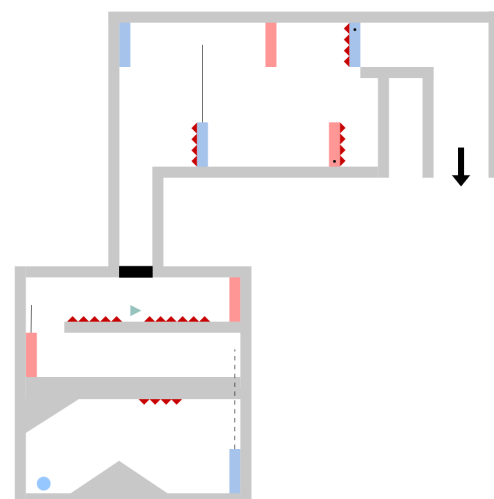


Figure: Level 1 Part 2 Sketch

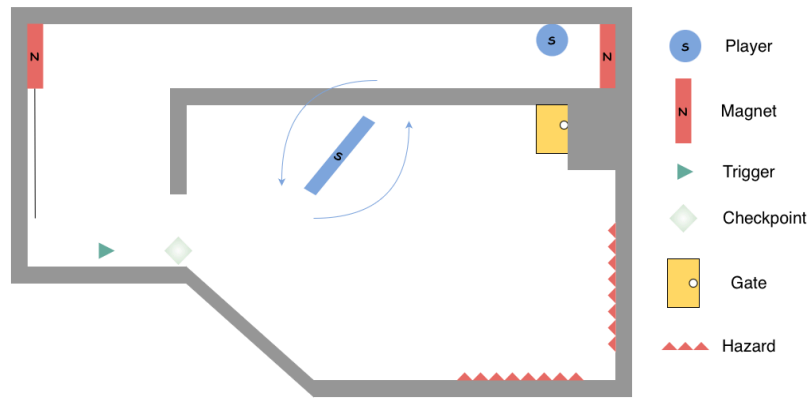


Figure: Level 1 Part 3 Sketch

Implemented:

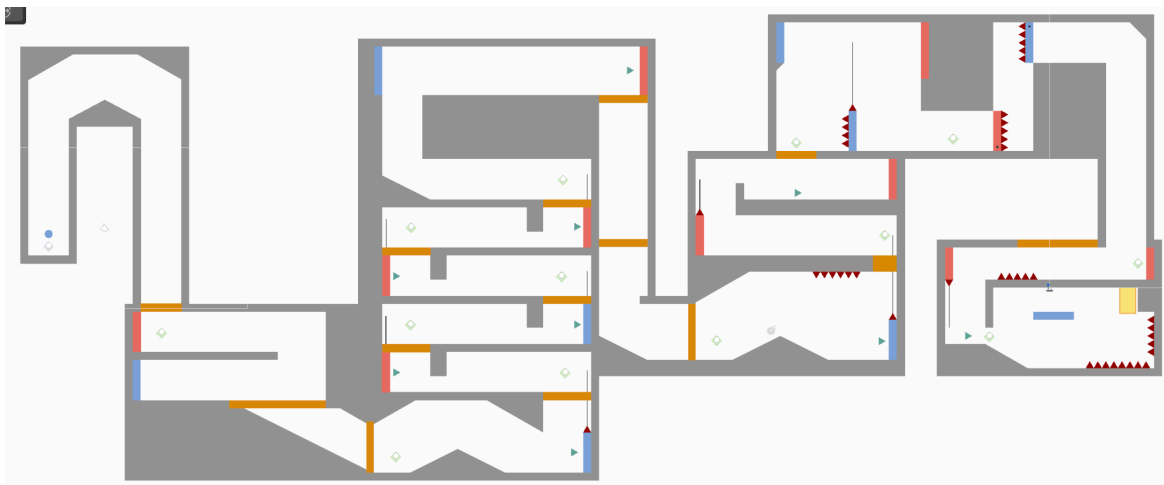


Figure: Level 1

Level 1 serves as a bridge between the tutorial and the more challenging levels. The first section reinforces the core mechanics of gravity switching and magnetic polarity control in a safe, clearly structured environment. In the latter section, the difficulty increases slightly with simplified versions of the traps and platform components that will appear in later levels, allowing players to gain confidence with the controls before progressing further.

- Level 2
Sketch:

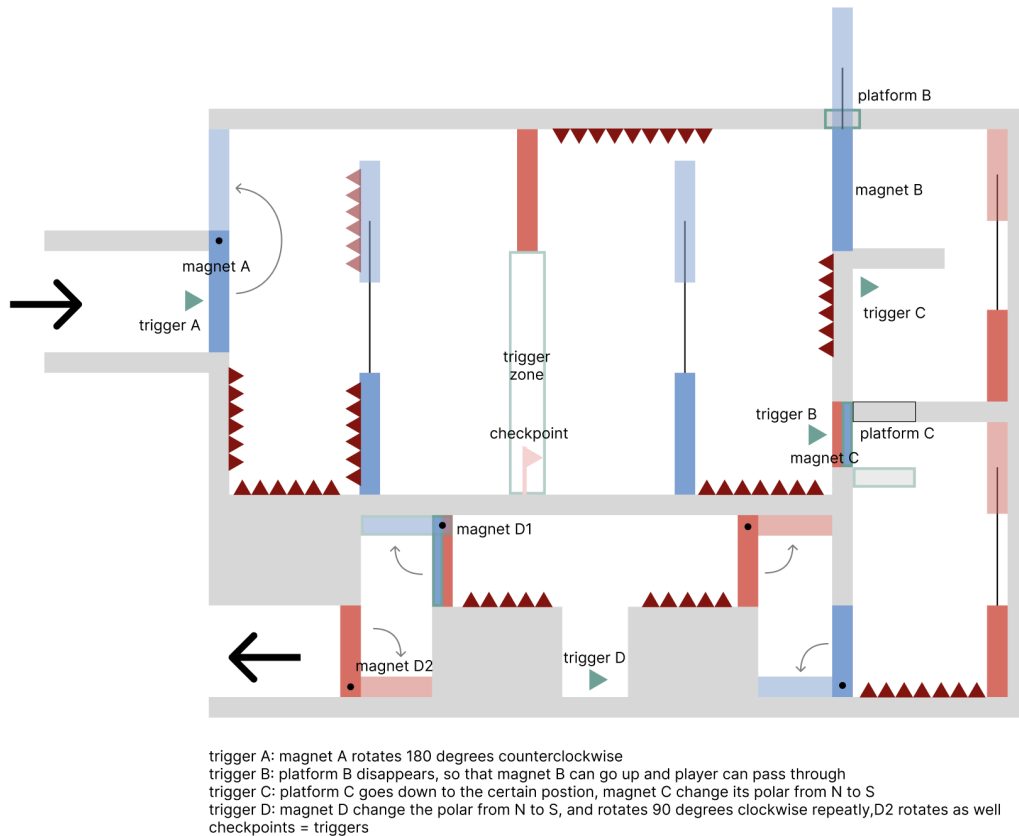


Figure: Level 2 Sketch

Implemented:

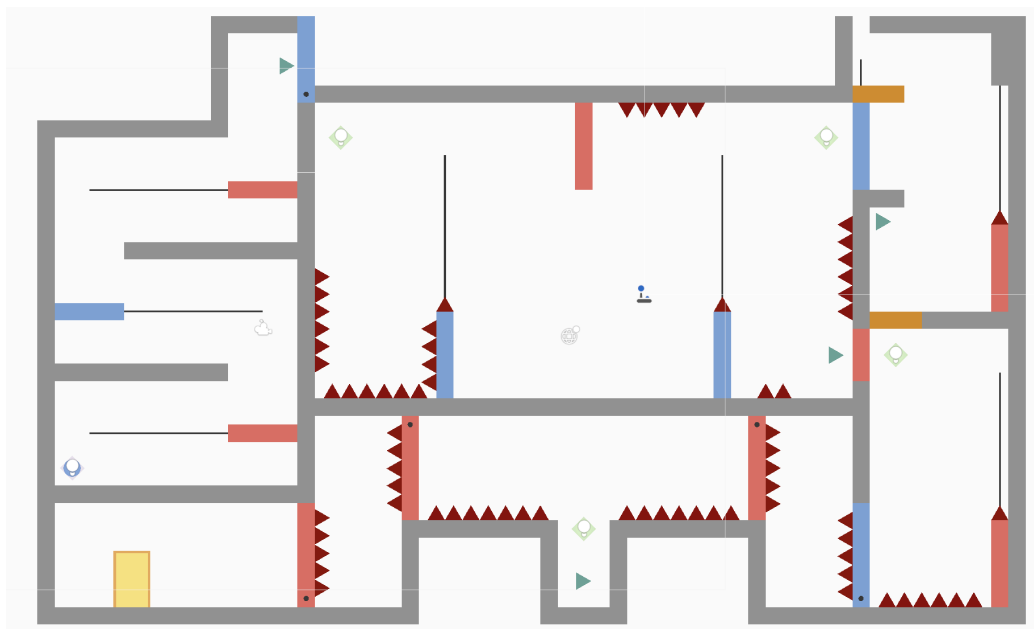


Figure: Level 2

Level 2 introduces stronger puzzle elements, requiring players to plan their route and time to switch the direction of gravity and magnetic polarity carefully. The first section is easy, allowing players to build confidence before facing more complex challenges. Many of the later challenges expand on concepts hinted at in Level 1, making the

solutions discoverable even as complexity increases. This level also includes simpler versions of platform components from Level 3, helping players prepare for the upcoming level.

- Level 3
Sketch:

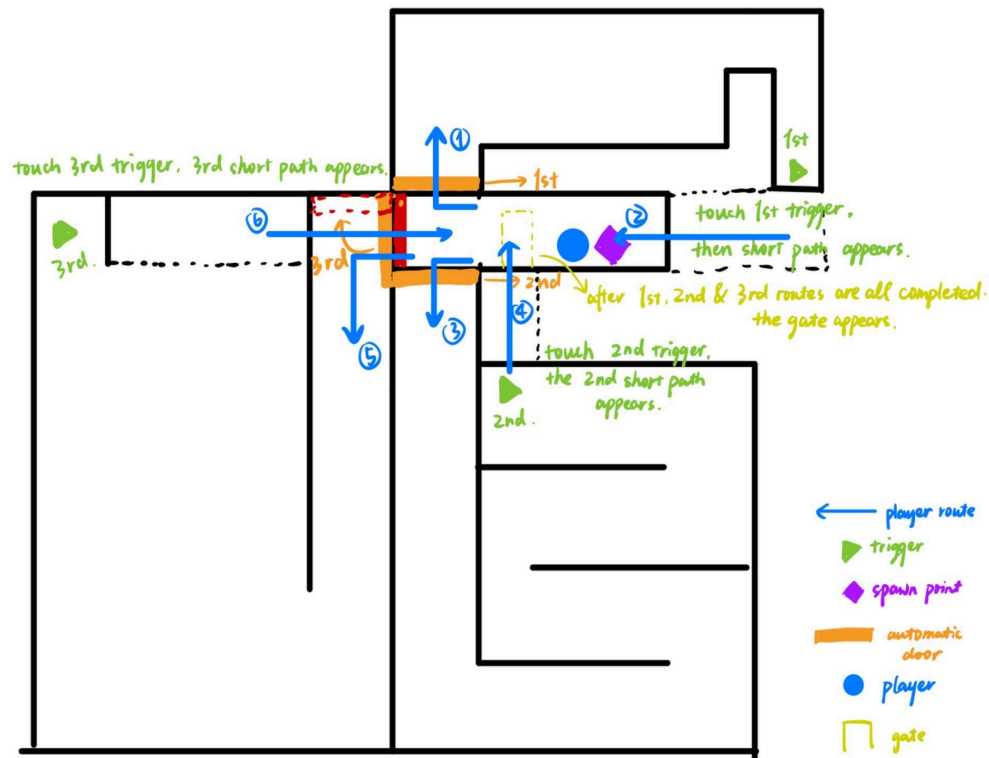


Figure: Level 3 Layout

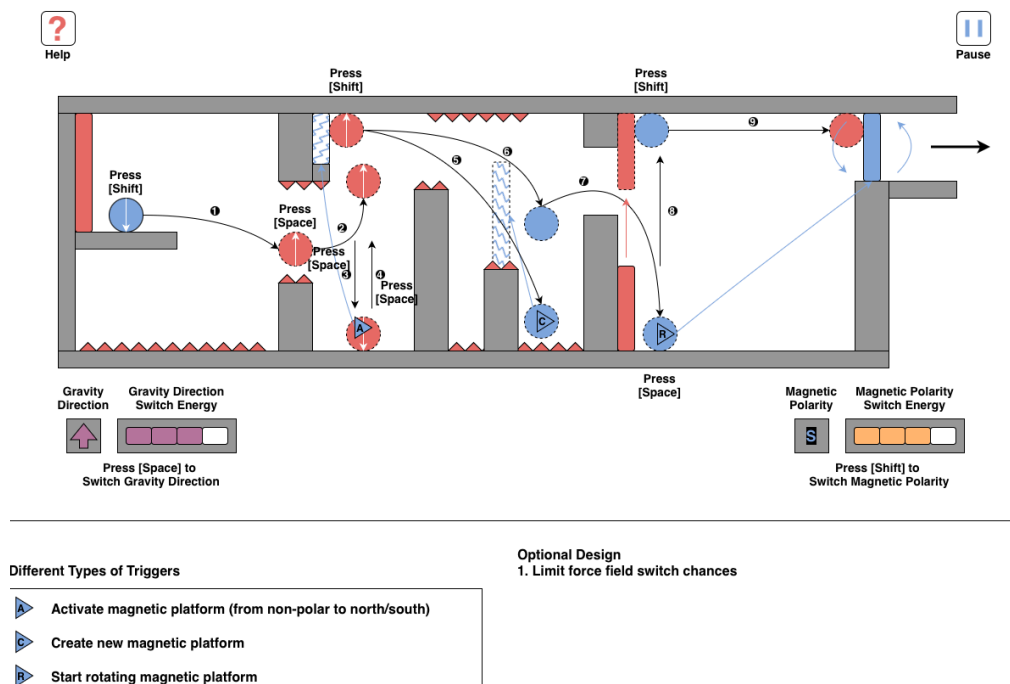


Figure: Level 3 Part 1 Sketch

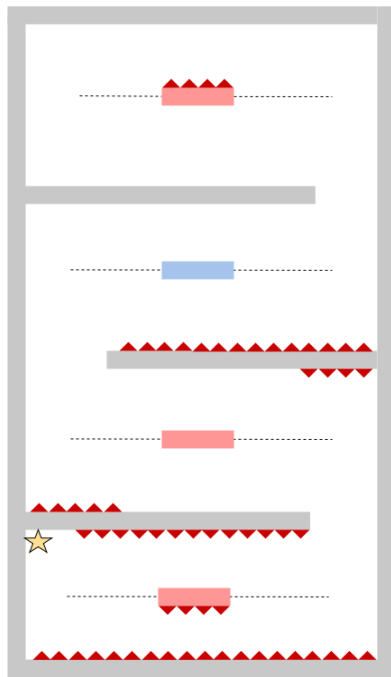


Figure: Level 3 Part 2 Sketch
Implemented:

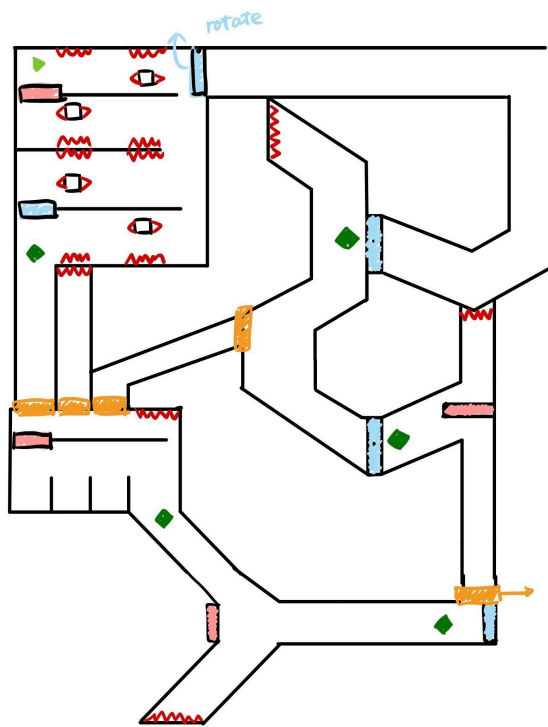


Figure: Level 3 Part 3 Sketch
Implemented:

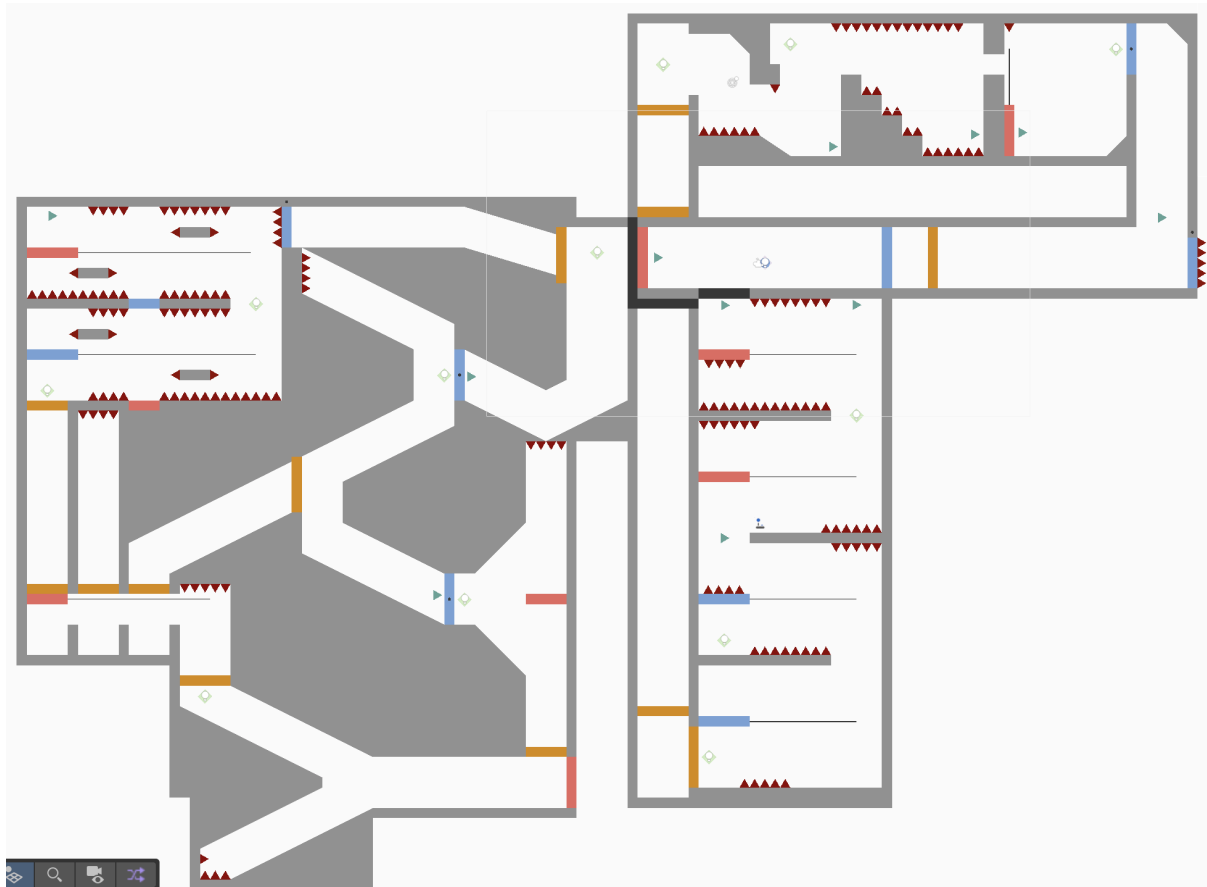


Figure: Level 3

Level 3 is the most challenging stage, requiring full mastery of gravity and polarity manipulation as well as familiarity with all platform components. The level is divided into three sections. The first section emphasizes precise timing when switching gravity and magnetic polarity. The second section increases complexity while expanding on the idea introduced in Level 2, offering new interactions. The final section features a maze-like layout that requires careful navigation and presents the most demanding puzzle in the game, delivering a fresh and brain-teasing conclusion to the level progression.

Game UI - (Beta)

Main Menu

The main menu contains our game title “FLIP” followed by two navigation buttons:

- START: navigating player to tutorial level.
- LEVEL: navigating player to level selection page.



Figure: Main Menu

LEVEL Selection Page

- Back to menu: a button for returning to the main menu.
- Level: A single row of buttons for the tutorial and 3 levels of the game is displayed. After clicking each button, the page will be navigated to the corresponding scene. Note that each level will not be unlocked until the gate in that level is reached.



Figure: LEVEL Selection Page

Level Screen Layout

This is the interface for the tutorial and each game level. At the top left corner, there are two buttons:

- Help: Clicking this button shows game control and basic description.

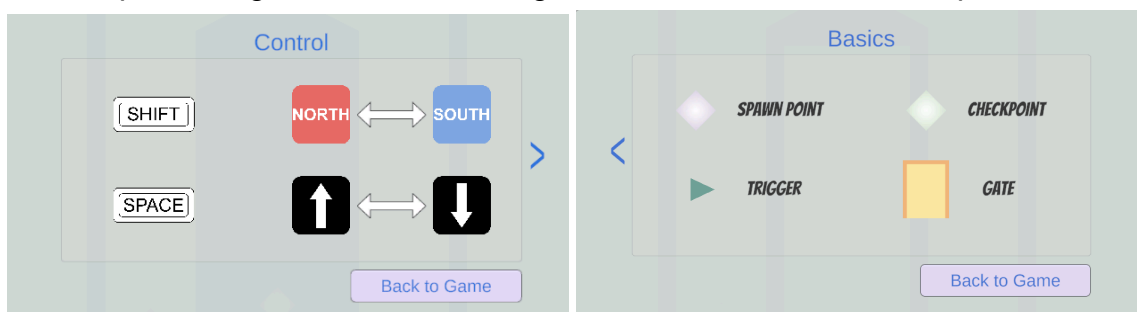


Figure: Help Page

- Pause: Clicking this button can pause the game. Then you can choose to go back to the game, choose respawns, restart the game, or go back to the main menu.



Figure: Pause Page

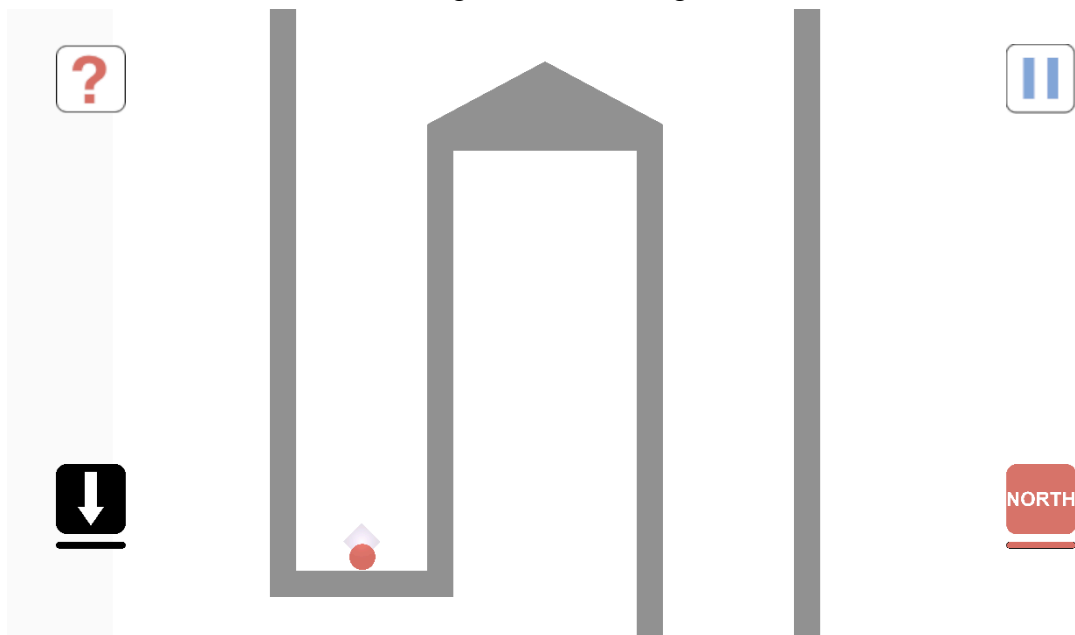


Figure: Tutorial and Level Page (same)

Level Complete Menu

Once the player reaches the gate, a Level Complete page is displayed.

- Restart: Clicking this button restarts the current level.
- Next Level: This button navigates the player to the next level or menu.



Figure: Level Complete Menu

HUD (Heads-Up Display)

- Magnetism indicator line: During gameplay, there is a line connecting each magnet wall with the player. If the line colour is read, it indicates a repulsive force between the magnet wall and the player, causing the player to move away from the wall. If the line is green, it indicates an attractive force, and the player will be pulled toward and stick to the magnet wall.

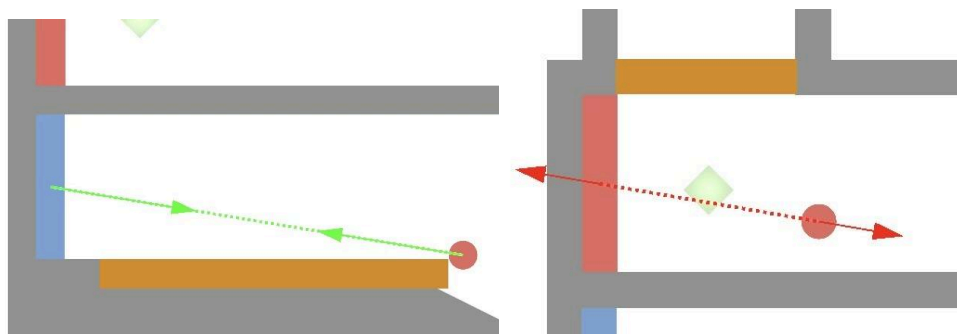


Figure: Magnetism Indicator Line

- Magnetism and gravity indicator:
For the gravity indicator, if the arrow points up, it means the global gravity is directed upward. If the arrow points down, it means the global gravity is directed downward. After the player presses the space bar to switch gravity, they must wait until the black charge bar below is full before switching again. For the magnetism indicator, if the indicator shows “SOUTH” and the colour is blue, it means the player’s current magnetic pole is the south pole. If the indicator shows “NORTH” and the colour is red, it means the player’s current magnetic pole is the north pole. Similarly, after the player presses the shift bar to switch magnetism, they must wait until the charge bar below is full before switching again.



Figure: Magnetism and Gravity Indicator

UI screen flowchart

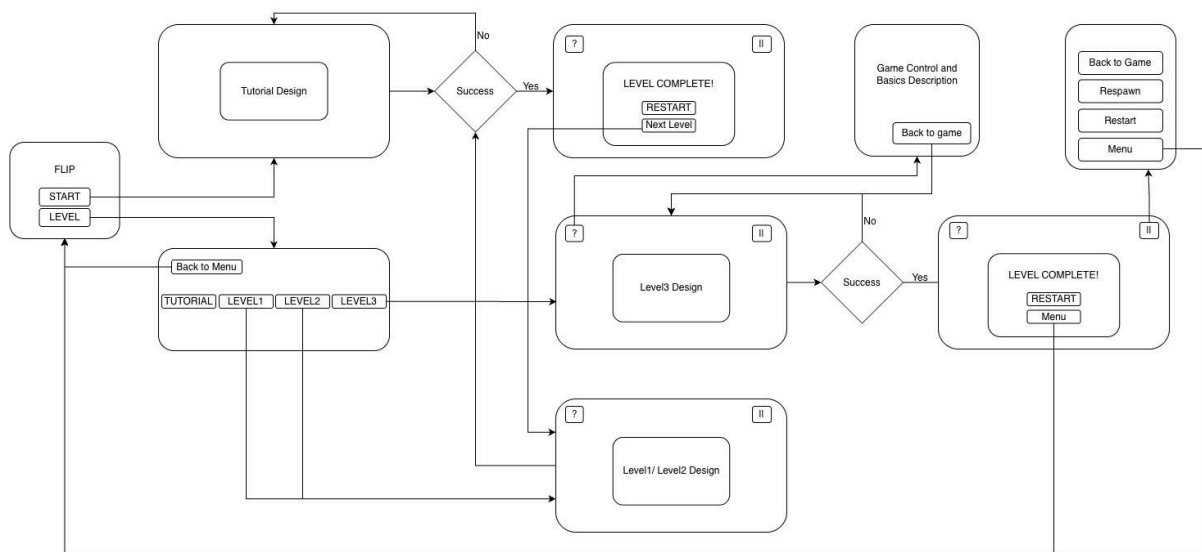


Figure: UI screen flowchart

Analytics

Methodology Used:

Data Collection:

Link to Data Uploading code:

1. [Death Spots Metric Script](#)
2. [Trigger Response Time Script](#)
3. [Player Action Switch Rate Script](#)
4. [Player Operation Efficiency Script](#)

Links to google form and google sheet:

1. [Death Spots Metric Google Sheets](#) & [Death Spots Metric Form](#)
2. [Trigger Response Time Form](#)
3. [Player Action Switch Rate Sheet](#) & [Player Action Switch Rate Form](#)
4. [Player Operation Efficiency](#)

Data Visualization:

For our metrics we have used Google Forms and Google sheets to visualize the data.

We retrieved the Death Spots data from google sheets and plotted it on top of the level map using python libraries like gspread and matplotlib.

Link to data visualizations folder:

[Death Spots Metric Folder](#)

Metrics

Metric #1: Death Spots (DS)

Description: *The Death Spots (DS) metric identifies the precise (x, y) coordinates within the game world where a player's death event occurs. This positional data is then aggregated and visualized as an overlay or heat map plotted directly onto the level design image.*

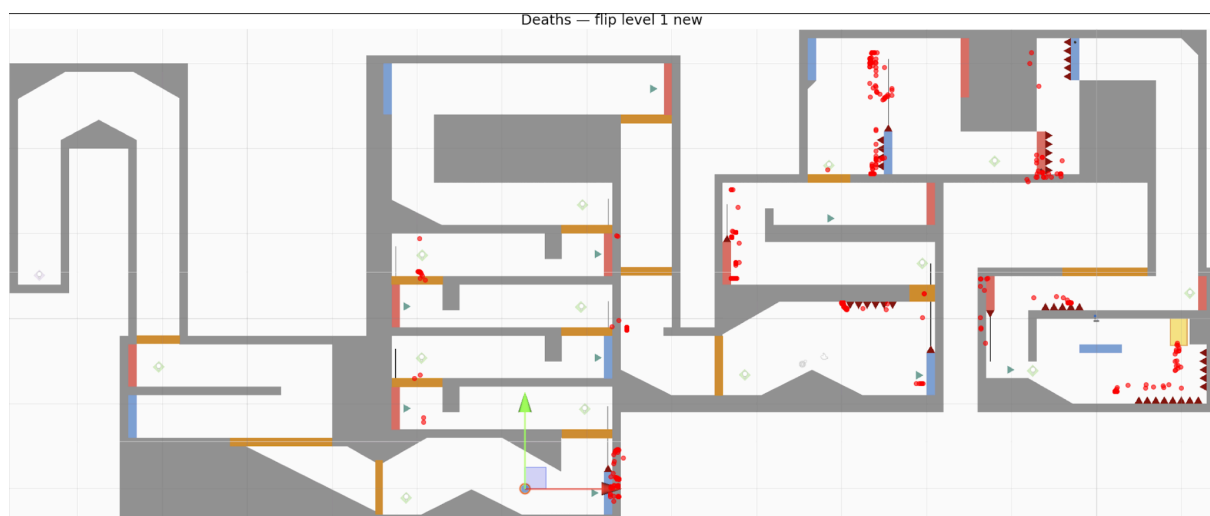
We retrieved the Death Spots data from google sheets and plotted it on top of the level map using python libraries like gspread and matplotlib.

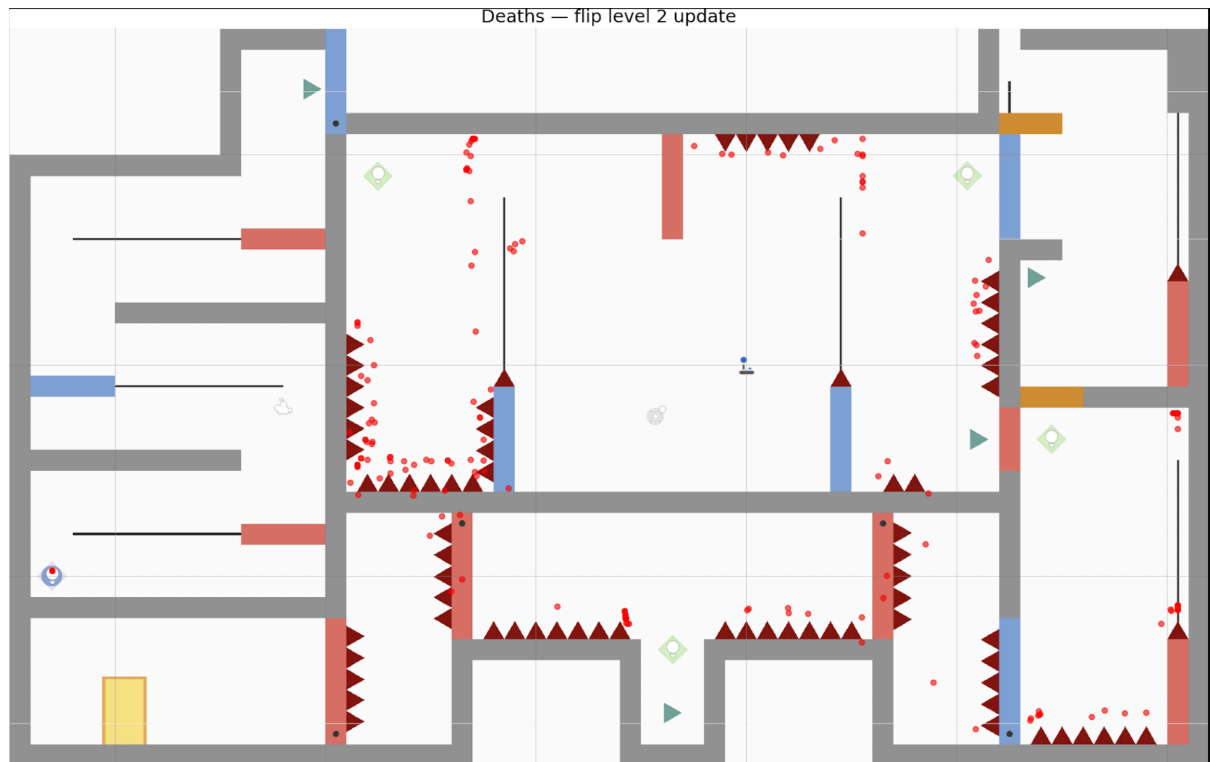
Data Collection: [Death Spots Metric Script](#)

Links to google form and google sheet: [Death Spot Metric Google Sheets](#) & [Death Spot Metric Form](#)

Data Visualization: [Death Spots Metric Code](#)

Beta Phase Metric #1 Visualizations:





Justification:

To pinpoint the exact geometry, enemy, or hazard causing player death, which is

often impossible to determine through check-point based metrics alone.

- **Identifies "Choke Points":** *It visually exposes specific locations (e.g., a tricky jump, a specific trap) that act as unexpected or overly difficult "choke points" for players.*
- **Validates/Invalidates Designer Intent:** *It helps answer questions like: Are players dying where we intended them to? or Are players dying due to unintended geometry issues (e.g., falling off a specific, poorly designed mechanic)?*

Interpretation:

*The visual density of the plotted points provides a direct measure of the lethal friction at a specific location. Analyzing the **shape and spread** of the death cluster helps determine the root cause.*

- **Possible Result 1: The "Tight Cluster" (Dense Death Spot)**
- **Possible Result 2: The "Wide Spread" (Diffuse Death Spot)**
- **Possible Result 3: The "Trail" (Linear Death Pattern)**

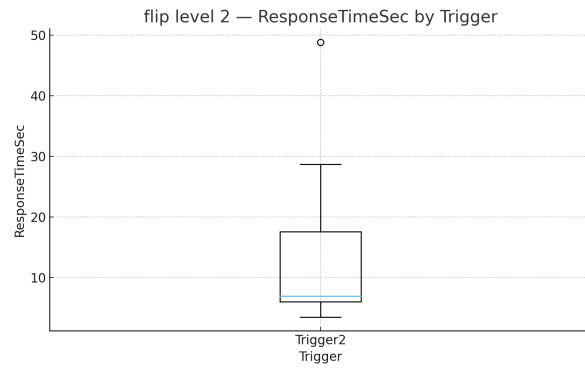
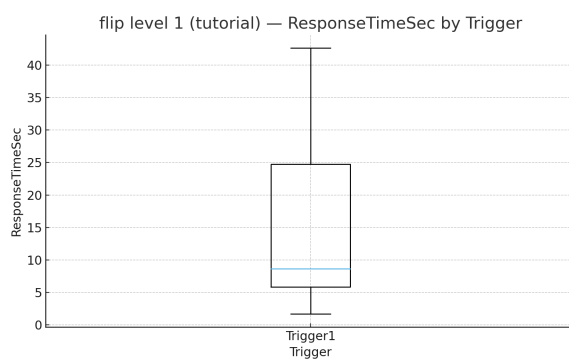
Hypothesis:

- *For the possible result 1, Introduce a short "buffer zone" before the hazard, widen the gap slightly, or add visual cues for optimal polarity shift timing.*
- *For the possible result 2, Simplify the control sequence (e.g., reduce the number of required polarity/gravity shifts), add a brief "rest platform" mid-sequence, or ensure the sequence clearly teaches a new combination of controls.*
- *For the possible result 3, Before the tricky segment, place a small platform that **deactivates all current polarity/gravity fields** and **forces the player to a complete stop**. The challenge begins only when the player intentionally initiates the required control sequence from rest.*

Metric #2: Trigger Response Time

Link: https://docs.google.com/forms/d/1-2VUyPFp9mV_KG4UA_t6G0ogbSFfZO9ydNh0V5mbZlk/edit#responses

Description: *The total time taken by the player between Trigger N and the corresponding mechanism it controls. This timer resets immediately after the player activates the next trigger.*



Justification:

This metric clearly shows how players interact with different triggers and mechanisms. By analyzing the time taken to complete the task, we can determine whether triggers are placed efficiently, and whether the difficulty is balanced for each stage or level. If players consistently take too long or too short, it might indicate that the trigger timing or mechanism activation is not intuitive or balanced, which leads to potential improvements.

Interpretation:

- *Possible result 1: long trigger response time. This means players are taking too much time to reach the mechanism after triggering. It may suggest that there is a poor trigger placement or high difficulty stage. This can indicate that the level design is too complex, or the player struggles to understand how to effectively use gravity and magnetism to reach the mechanism.*
- *Possible result 2: short trigger response time. This means players reach the mechanism too quickly. This may imply that the level is too easy, or the trigger-to-mechanism distance is too short. Another possibility is that players find a short path because sometimes polar magnetism is uncontrollable.*

Hypothesis:

List testable hypotheses based on the data you will collect. What could be the cause of variations in your data, and can this cause be modified in your game to yield different results?

- *For possible result 1, the level design is too complex. We can simplify level complexity by adding clear visual cues or positioning triggers closer to corresponding mechanisms.*
- *For possible result 2, we may add more obstacles between triggers and their corresponding mechanisms.*

Metric #3: Player Action Switch Rate

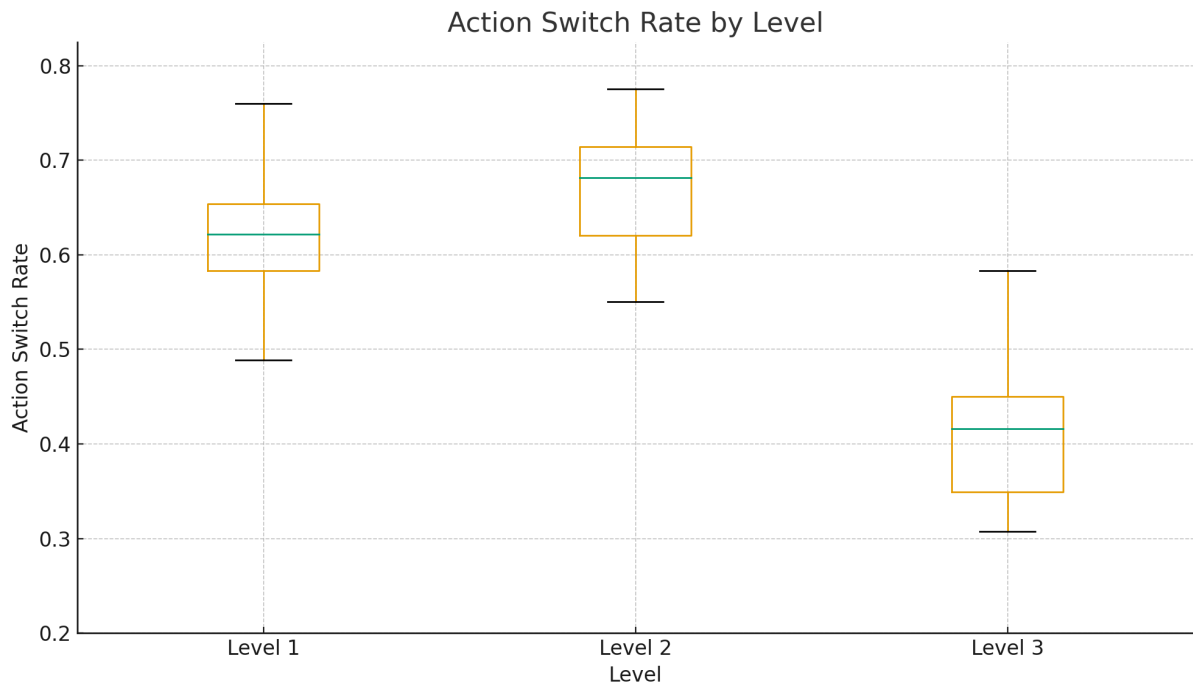
Link: <https://docs.google.com/forms/d/1UNOQsuxnaT-dVYEs7x5OaJ7zlyZBoRLJoNBh-5pqLjw/edit#responses>

Description:

- Given that the player has two distinct actions within the 'Flip' mechanic (pressing [Space] to shift gravity direction, and pressing [Shift] to shift magnetism polarity), we can analyze player's preference for the following playing strategies:

1. Single-Mechanic Strategy: Navigate through a part of the maze by utilizing only one force field modification mechanic
 2. Dual-Mechanic Collaboration Strategy: Navigate through a part of the maze by the maze by collaboratively using the two different force field switching mechanics
- We collect the player's operations from the start of the level until its completion and consider this sequence of operation as the user action sequence. The length of this sequence is denoted by N , and the i -th action is represented by Op_i . We define the Action Switch Rate as follows:

- Action Switch Rate = $\frac{\sum_{i=1}^{N-1} I(Op_i \neq Op_{i+1})}{N}$, where $I(\cdot)$ is the indicator function, which equals 1 if adjacent actions are different, and 0 otherwise



Justification:

The Player Action Switch Rate serves as a metric to determine whether the mainstream player strategy aligns with our intended level design.

1. If our objective is to minimize player confusion by designing levels that can be solved by focusing on only one force field at a time (gravity or magnetism), and even to encourage players to concentrate more on the timing of single-field switching, then we should aim for a low Player Action Switch Rate
2. If we seek to combine the two different force fields switching mechanics more organically, thereby requiring players to actively consider the collaboration of both fields to solve the puzzle, we will necessitate a high Player Action Switch Rate to confirm the effectiveness of our design

Interpretation:

- Players who favor Strategy 1 mentioned above should exhibit a low Action Switch Rate
- Players who favor Strategy 2 mentioned above should exhibit a high Action Switch Rate
- If the data variance is high, it suggests that different users prefer different strategies
- If the data variance is low, it suggests that most users prefer the same strategies

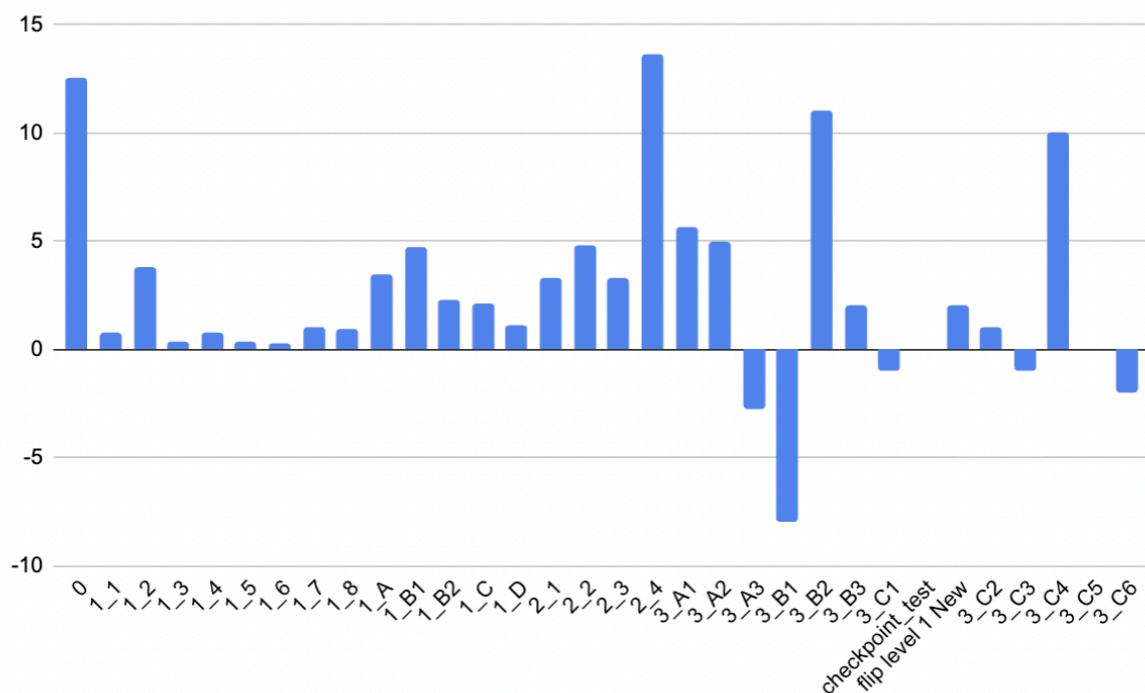
Hypothesis:

Our current level design is intended to minimize player confusion. In our internal testing, we confirmed that most puzzles can be solved by modifying only a single force field. Therefore, we anticipate that the majority of players will exhibit a low Action Switch Rate.

Metric #4: Player Operation Efficiency

Link: <https://docs.google.com/forms/d/1bYAn1ZaUwffPutYLoEyktF7P6ipTj076vApvDSQCJII/edit>

Description: Count the total number of player operations that occur between Checkpoint N and Checkpoint N+1. Then compare this number to the ideal number of operations, which is given by us (map designer). The counter resets immediately each time the player reaches the next checkpoint.



Justification:

The difference between the number of player operations and the number of ideal operations can clearly tell us how deep the player understands the level. If the player operates too

much than we expect to find a way out, it may reflect the difficulty of the level is too high or the instruction is not clear enough. If many players operate less than we expect, i.e. there is an obvious short path, we may decide whether to keep the short path as a bonus or change the design to avoid the short path.

Interpretation:

- *Possible Result 1: Low difference (close to ideal), with only a few below the ideal operation count: Players understand the level design logic well and can efficiently use the core mechanics.*
- *Possible Result 2: High difference (more than expectation): Players are likely struggling with navigation or mechanics comprehension and operation. The cue to lead them out may be unobvious or lacking. Or, the level design may be unintuitive or too difficult for this stage.*
- *Possible Result 3: A portion of players perform fewer operations than the ideal operation count: Players can easily find a short path to complete this segment of level.*

Hypothesis:

List testable hypotheses based on the data you will collect. What could be the cause of variations in your data, and can this cause be modified in your game to yield different results?

- *For Possible Result 1: The challenge balance is appropriate. The level design is effective.*
- *For Possible Result 2: We may add more obvious cues in the level design to show the way out. Or, we may change the unintuitive parts and simplify the most difficult part.*
- *For Possible Result 3: We may decide whether to keep this short path as a bonus for players or delete this short path as a defect of the level design.*

Weekly Prototype Descriptions

Week-7:

- *Discussed how to combine magnetism and gravity in gameplay.*
- *Explored level design structures to maintain simple controls and enjoyable gameplay.*
- *Wrote and combined the initial scripts.*
- *Completed team role assignments*

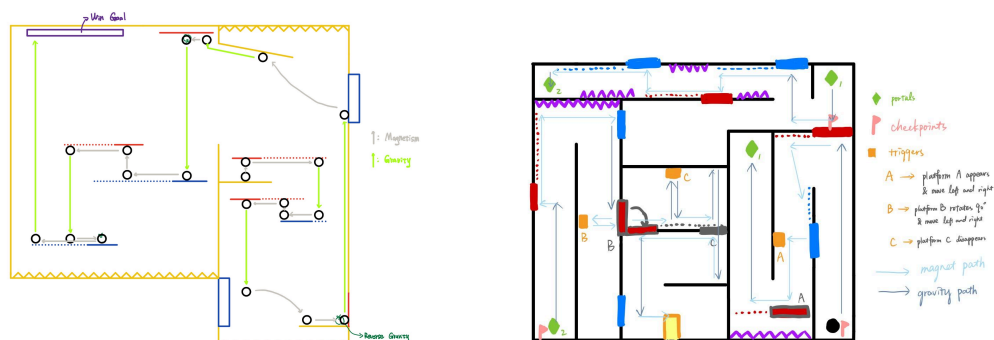


Figure: Early stage discussion of different level design ideas

Week-8:

- Collaboratively designed and implemented the tutorial level.
- Finalized the documentation for the first prototype progress check.
- Completed and tested all scripts related to the tutorial level.

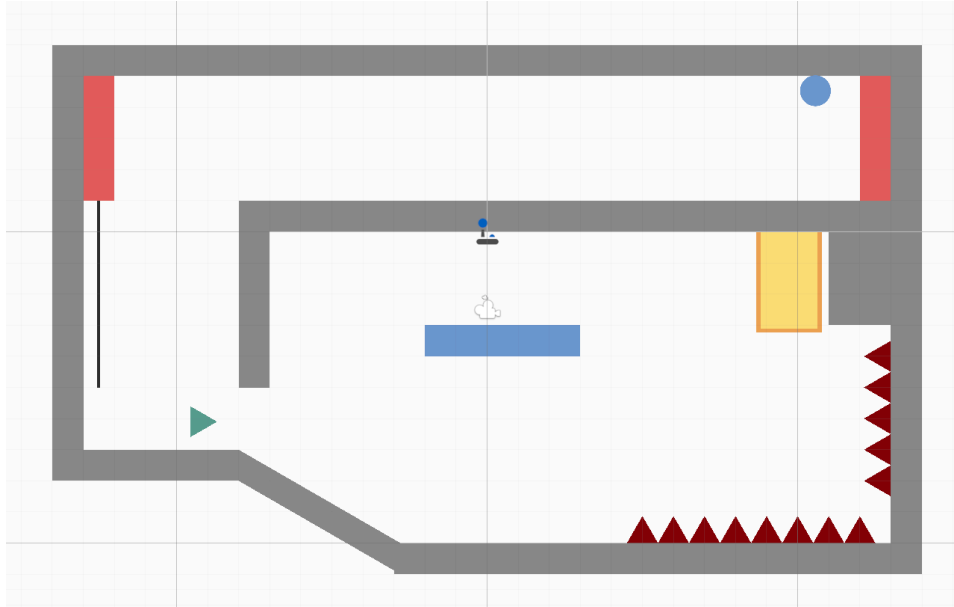


Figure: Complete Tutorial level

Week-9:

- *Improve the tutorial level based on the feedback from the playtest.*
- *Collaboratively designed and implemented the Level 2.*
- *Figure out the appropriate analytics metrics and complete the code for data collection and visualization.*
- *Finalized the documentation for the Alpha Milestone Build*
- *Made new UI design and The project successfully runs on both iOS and Android platforms, with the touch-optimized UI implementation.*

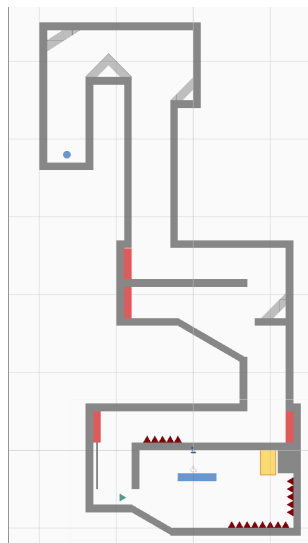


Figure: Complete Improved Tutorial level

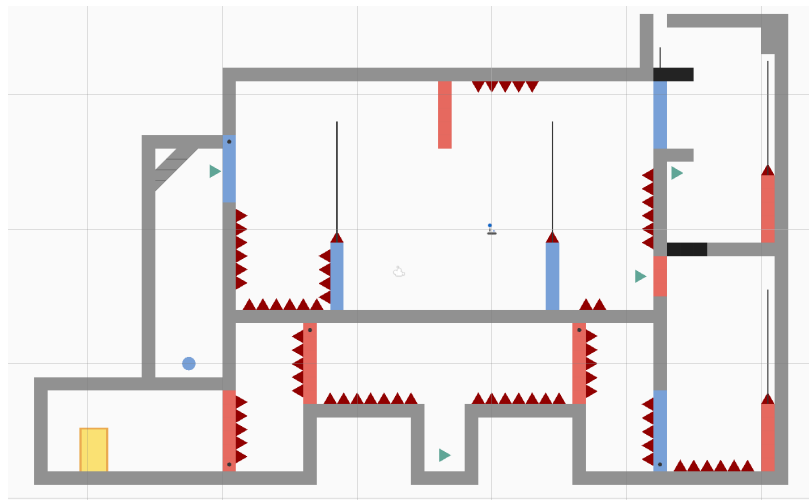


Figure: Complete Level 2

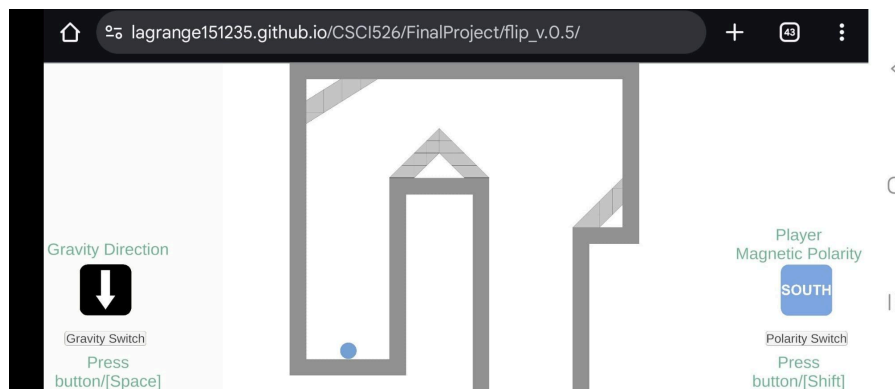


Figure: Game running on mobile devices

Week-10:

- Team Flip focused on brainstorming and experimenting with new mechanics and gameplay and new map design this week.
- Discuss and Brainstorming how to adjust the in-game tutorials based on the feedback collected from the survey.
- Brainstorm new mechanics, UI designs, and animation effects, and explore how to apply them in the game.
- Make adjustments to Level 2 using different approaches based on the feedback from the playtest.

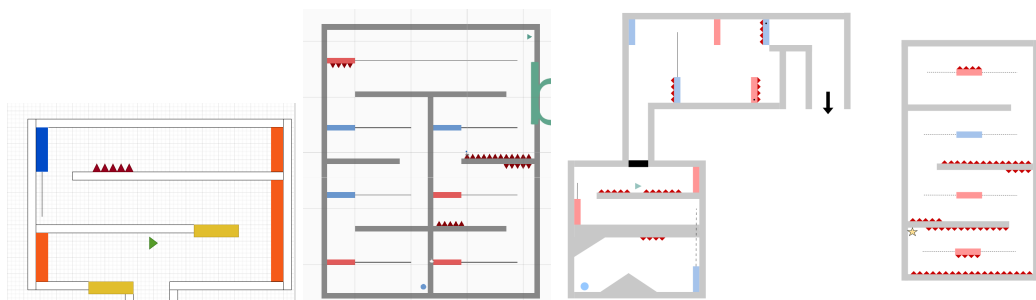


Figure: Different New Map Design

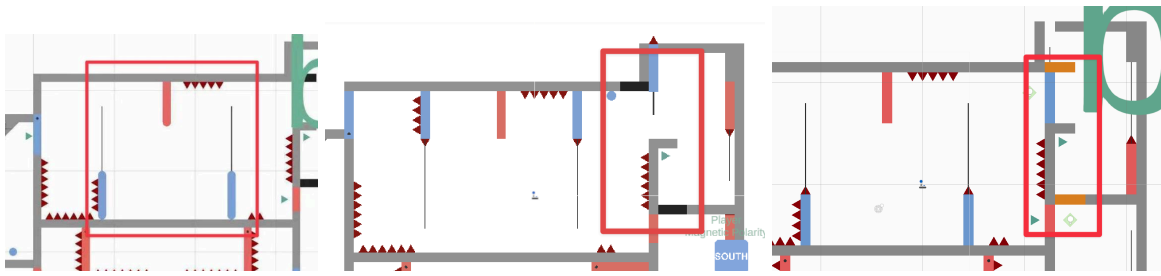


Figure: Testing Different Version of Level-2

Week-11:

- Build and update the new ideas and concepts from previous discussions into the existing levels.
- Create Main Menu and connect each level to the menu page and level selection page.
- Build new tutorial level and adding new animation for tutorial
- Design new Level 1 based on feedback from survey results
- Design and implement first version of Level 3
- Design and implement Magnetism indicator line

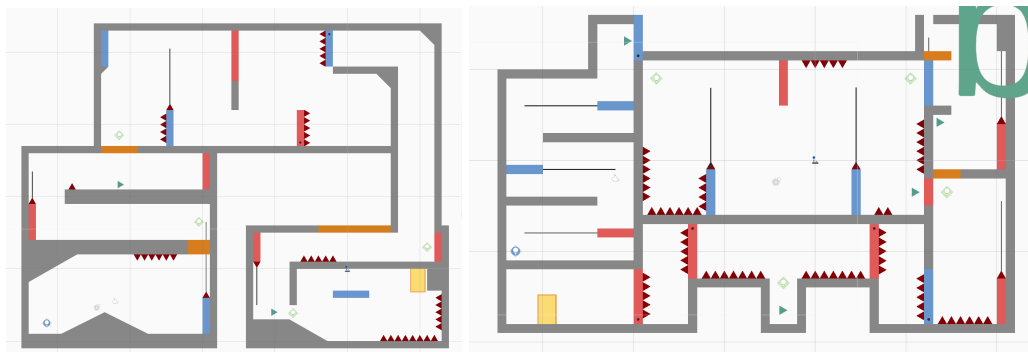


Figure: New Map Design for level-1 and level-2

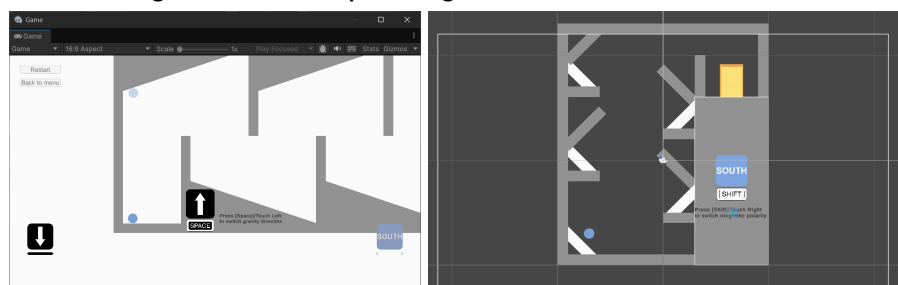


Figure: New Tutorial Methods and Design New Tutorial Levels.

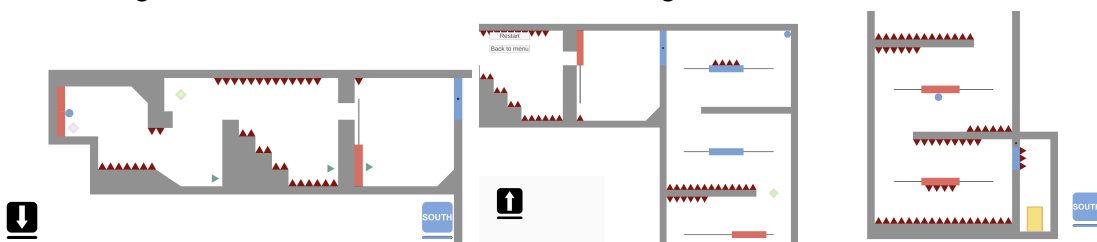


Figure: Implemented first version of level-3

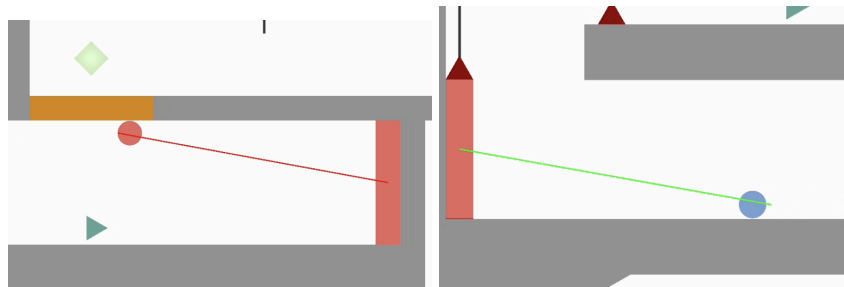


Figure: Implemented Magnetism indicator line

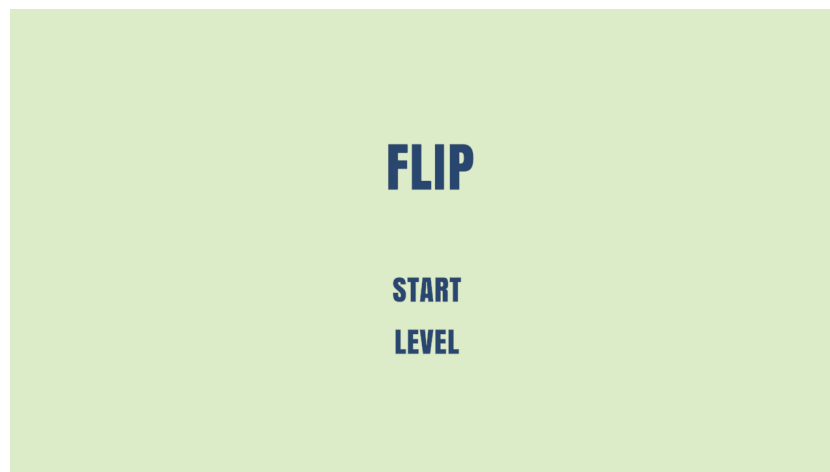


Figure: Create Main Menu

Week-12:

- *Redesign Level 1 based on feedback from weekly playtest*
- *Design and implement the final version of Level 3*
- Add new Help panel and Pause panel
- Modify the level selection page
- Adjust the magnetic animation effect with the redesigned magnetism indicator line
- Implement all analytics metrics

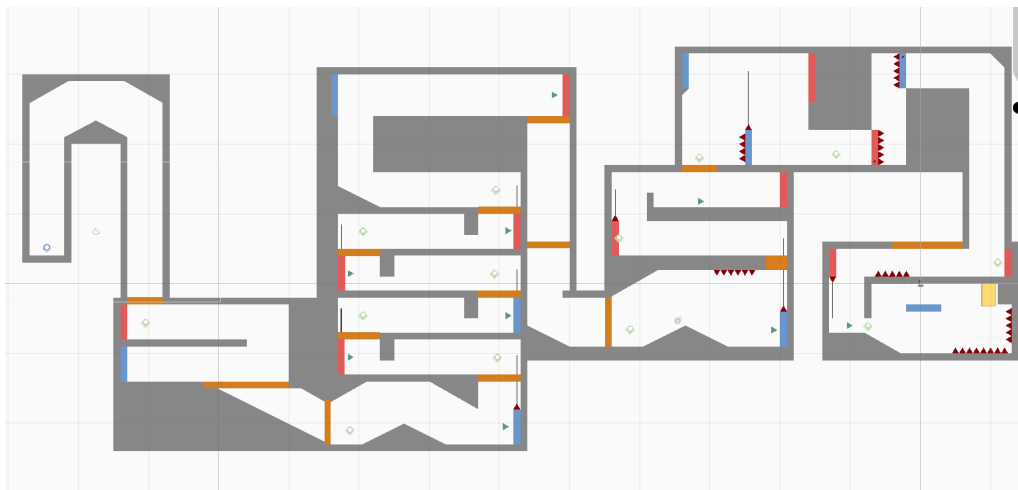


Figure: New Level one with new tutorialization

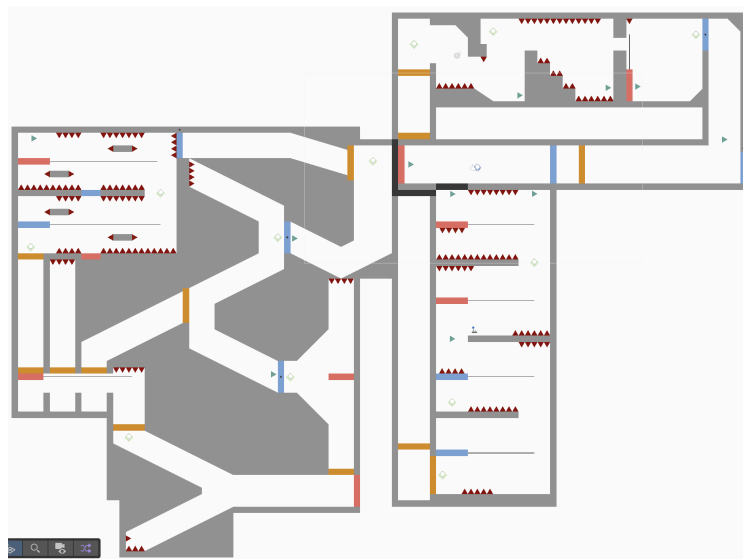


Figure: New level three with maze layout

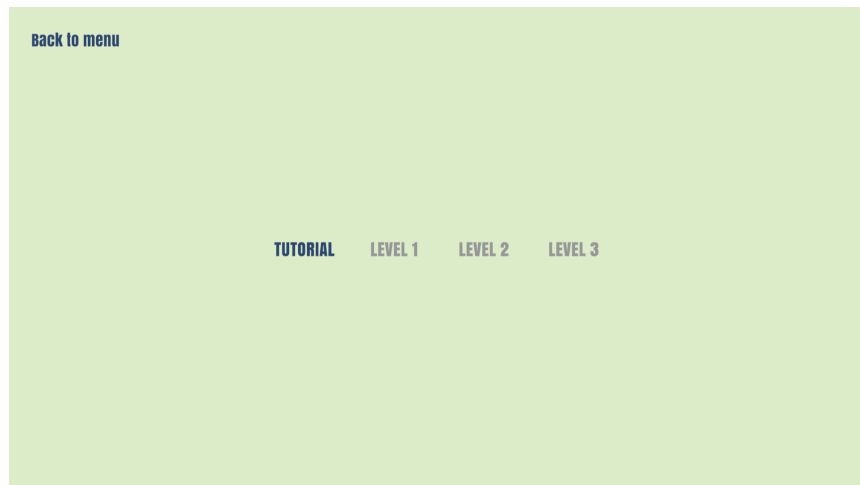


Figure: New Level Selection UI

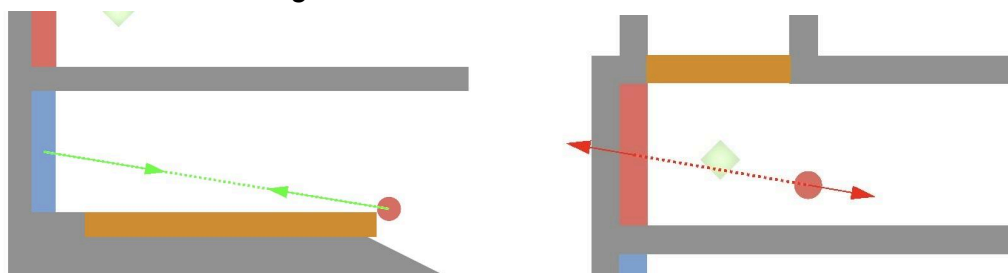


Figure: Implement New Design of Magnetism Indicator Line

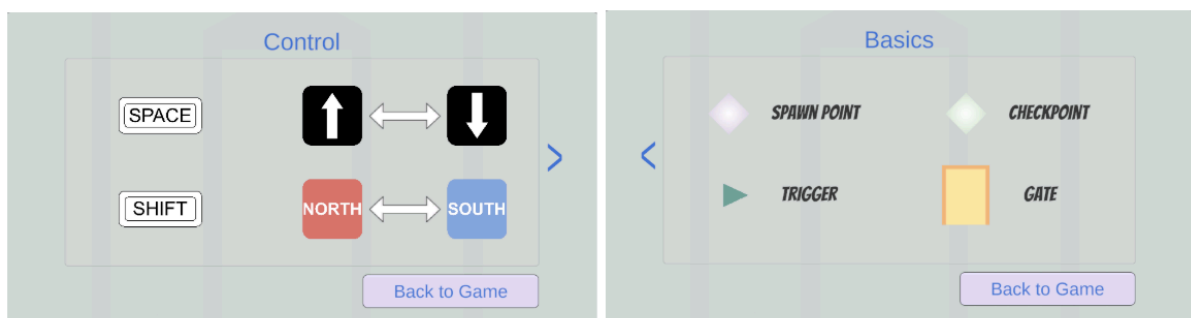


Figure: New Help Panel



Figure: New Pause Panel

Week-13:

- Add Quick Respawn Button - Press "R" to respawn immediately or wait to respawn
- Implementing an Adaptive "Safe Mode." - Add the power-up to eliminate hazard when died 5 times
- Implement key binding modification function

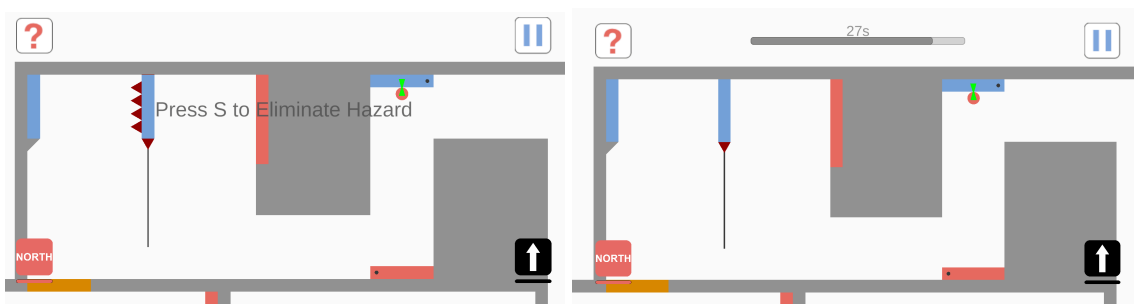


Figure: New Key Binding Panel

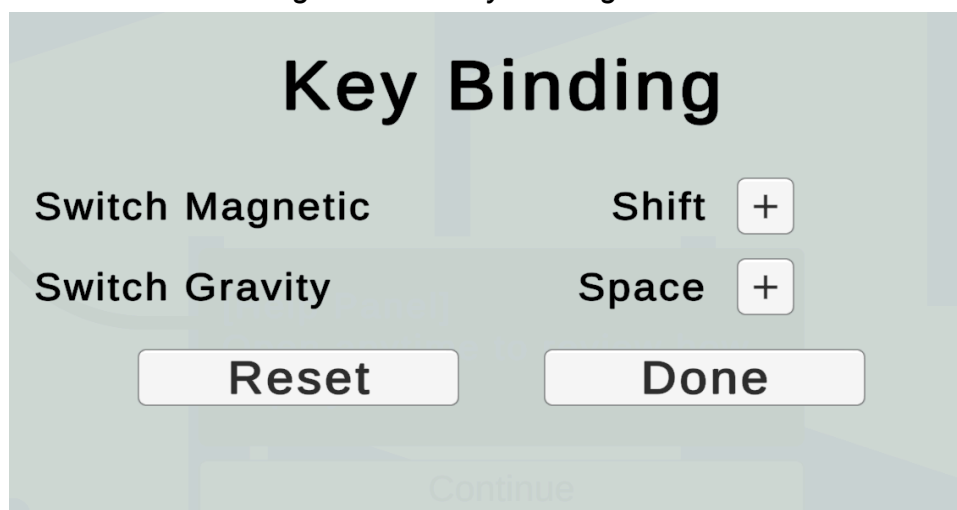


Figure: New Key Binding Panel

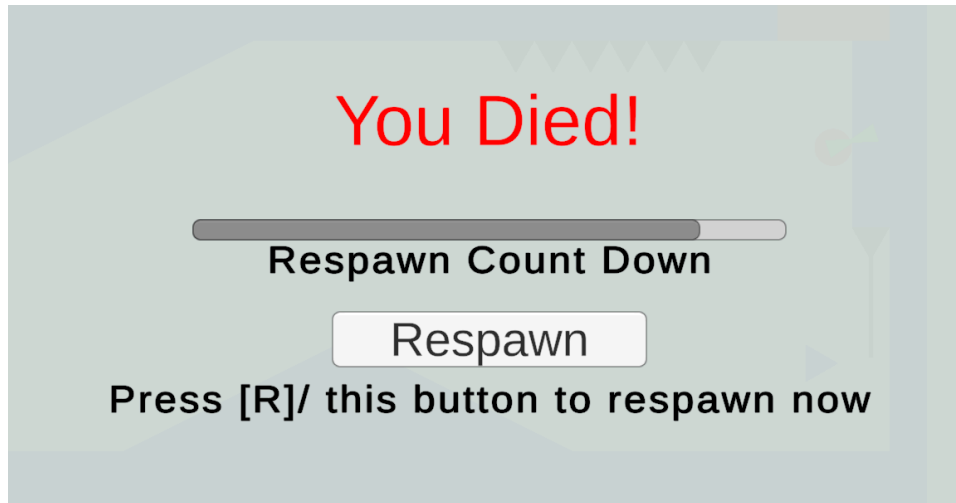


Figure: New Key Binding Panel

Week-14:

- Try to implement Arts and Sounds in Game
- Fixed all Autogate bugs in 3 levels
- Let images and texts sync with key binding

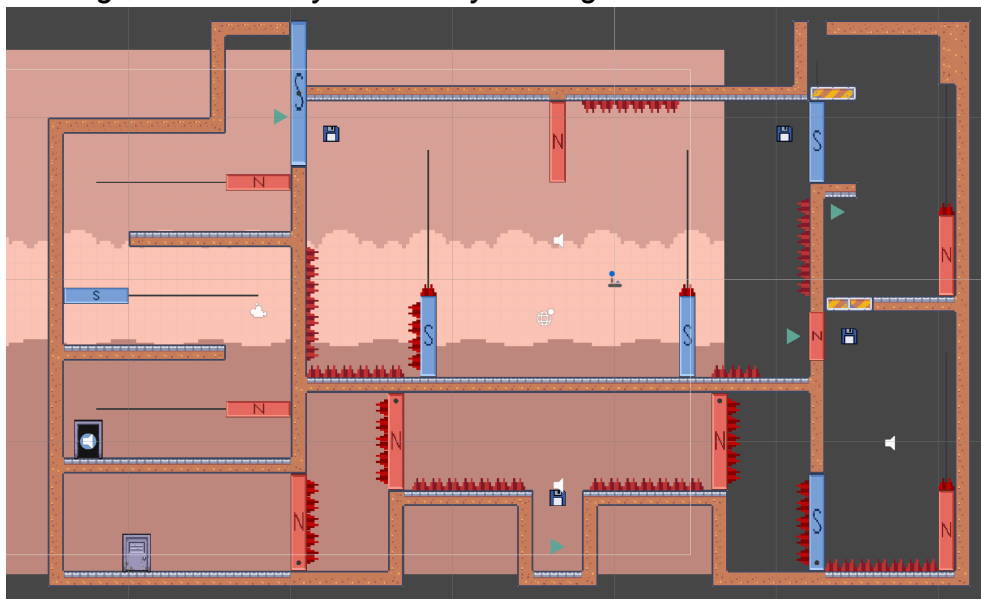


Figure: Arts Implementation



Figure: Arts Implementation

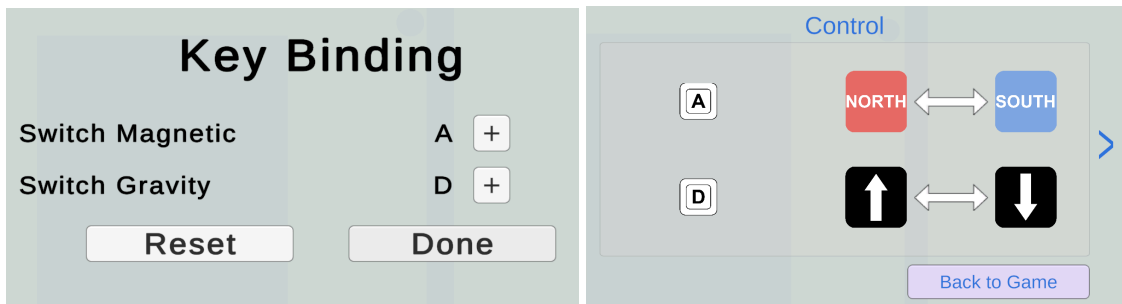


Figure: Key Binding sync on UI

Week-15:

- Add introduction UI at gravity tutorial
- Add Background music and sound effect
- New Main Menu and Level Selection Page
- New Level One with Art

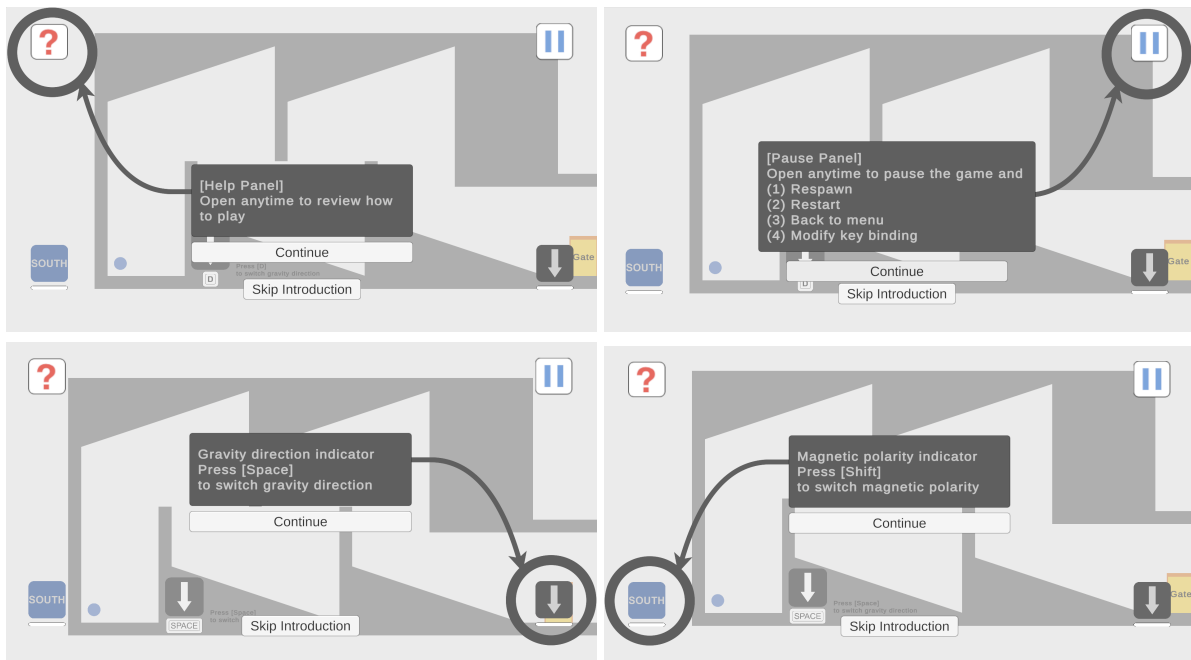


Figure: Introduction UI



Figure: New Main Menu and Level Selection Page

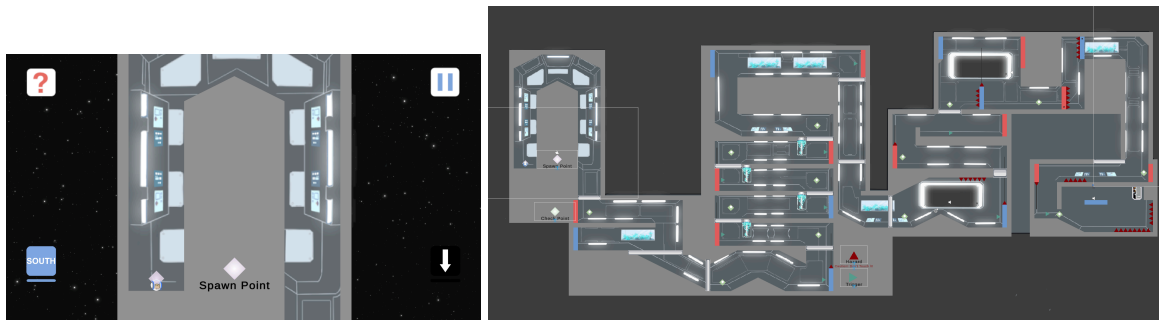


Figure: New Level One with Art

Weekly Progress

Week Number	Team Member	
Week 7	Xiaoyu Ke	Sherry Wang
	Level design & upload Camera Control and platform mechanisms code https://github.com/CSCI-526/main-flip/commit/bedbf77ca373cdc52b41566d9b37cab68ae946fc	Platform Movement & Trigger Interaction https://github.com/CSCI-526/main-flip/commit/7a9bc23a9bafa83153074b47058e9b822c47f890
	Xinlu Xu	Yang Li
	Cleaned and uploaded paired prototype code & Anti-gravity effect implemented https://github.com/CSCI-526/main-flip/commit/caa42e4777e830c8e9cc416440a097b749e09375	Level design & Cleaned and uploaded paired prototype code https://github.com/CSCI-526/main-flip/commit/8e2678948ebcc6e01a7156c8b2de445b0c856713
	Jiayi Zhang	Aryan Avinash Sawant
	Uploaded magnetism implementation https://github.com/CSCI-526/main-flip/commit/aad2e3005d82f0cb1de8e32e85b576fc56decde5	Uploaded Player Magnetism and Player Control Script https://github.com/CSCI-526/main-flip/commit/cc3cf31b1f318a7727b8ba712f34b723b5a503f1
Week 8	Xiaoyu Ke	Sherry Wang
	Hold meeting, Schedule plan, Debugging, Level design, Camera Control	UI elements https://github.com/CSCI-526/main-flip/commit/8a4abf5ca

	https://github.com/CSCI-526/main-flip/commit/6daf0ad976c3a06d8d396384df453aab002b2e31	e08ad6bf14053a626bc795733727b89
	Xinlu Xu	Yang Li
	Anti-gravity modified & Light follow effect https://github.com/CSCI-526/main-flip/commit/e3d151696c7d224d3bfdb1603776d8d39641f30	Level design implemented https://github.com/CSCI-526/main-flip/commit/9e7c2b079d16b3612d330d23d50fe52e8f137395
	Jiayi Zhang	Aryan Avinash Sawant
	Upgrade magnetism implementation & Add friction implementation https://github.com/CSCI-526/main-flip/commit/f8fd1b9288f8c4dc9b5002a96c97e818af169737	Add Restart Button https://github.com/CSCI-526/main-flip/commit/1f1289888b3c459d64ee74f83ca2c489eaa17297
Week 9	Xiaoyu Ke	Sherry Wang
	New Map Design for Level One https://github.com/CSCI-526/main-flip/commit/30b92d4872b54cac1d94eed94ec8db9ccb88c060	Enhanced Game UI Panel, Implement metrics analytics https://github.com/CSCI-526/main-flip/commit/f1709b39a786cc3d532d51fded33e5aff99dae79
	Xinlu Xu	Yang Li
	Add bounce effect & Matrix analytics https://github.com/CSCI-526/main-flip/commit/3c77543605633ce9f5ec440a3cd6a8fcd186123	Design and implement Level 2 https://github.com/CSCI-526/main-flip/commit/3ee1a0cc84a1a81c37e01faebc3a7e5d33d2a684
	Jiayi Zhang	Aryan Avinash Sawant
	Add Dynamic Camera Control & Force Field Indicators https://github.com/CSCI-526/main-flip/commit/8dd33887775821ee1b3a98178d81960d2c485046	Added Main Menu https://github.com/CSCI-526/main-flip/commit/5d351f7b5152e690ddb0f53b0040edf26cee184
Week 10	Xiaoyu Ke	Sherry Wang

	New map design and Hold meeting for brainstorming https://github.com/CSCI-526/main-flip/commit/9fe016733435ccd9fb992de09ce8346366f98eda	https://github.com/CSCI-526/main-flip/commit/4373c7dda7bb939ffadd6b7a5a4c635073fb2ecd
	Xinlu Xu	Yang Li
	Added main menu and connect it to responding levels https://github.com/CSCI-526/main-flip/commit/de0b6fa208fe41bee38942c66822d6e7e951ddd7	Make adjustments to Level 2 https://github.com/CSCI-526/main-flip/commit/5a960e2880a64cbe041ffdfa3da12237e2806ce
	Jiayi Zhang	Aryan Avinash Sawant
Week 11	Implement Force Field Recharge Mechanics https://github.com/CSCI-526/main-flip/commit/0fdcfd8f76b293b6bb8868df6918e112c1e305	Implemented sheets for data collection and integrating it with app script 📄 Death https://github.com/CSCI-526/main-flip/commit/7129907148a1cc988cbdd4615001618ea5c4c0d0
	Xiaoyu Ke	Sherry Wang
	Build new level one and try different ways of tutorialization https://github.com/CSCI-526/main-flip/commit/866ab6d5c262daa508973db0c59101ca74095a87	added menu button https://github.com/CSCI-526/main-flip/commit/d0ca49aef98d1b5435d88de7ed3c95498d5235b2
	Xinlu Xu	Yang Li
	Made adjustments to main menu https://github.com/CSCI-526/main-flip/commit/13bd5409c4bb5d959f3834772ec9aab858407e8f	Designed and implemented the first version of Level 3 and new Level 1 https://github.com/CSCI-526/main-flip/commit/c295d86b499268f7c081ad0e2465061712351831
	Jiayi Zhang	Aryan Avinash Sawant
	Implement New Tutorial https://github.com/CSCI-526/main-flip/commit/e5b5312fbce919041aba6a098db1bf1f7186cc1c	Worked on New Metric: Death Spots https://github.com/CSCI-526/main-flip/commit/6b314b4af73849ac7d3cdec0bb51d53

		19db4c45
Week 12	Xiaoyu Ke	Sherry Wang
	Build a new level one with more tutorialization & debug for camera and trigger https://github.com/CSCI-526/main-flip/commit/68f48857c06d393fd8566f9b0902dbdb36619eef	Implemented the 4th metric https://github.com/CSCI-526/main-flip/commit/8922edb736821f0384789977b095d7657a031d15
	Xinlu Xu	Yang Li
	Implemented Player Action Switch Rate Metric https://github.com/CSCI-526/main-flip/commit/39937eaa6b5b03c611a937acb948945f6118b583	Designed and implemented final version of Level 3 & Added "Help" and "Pause" panel https://github.com/CSCI-526/main-flip/commit/89047511433cd979a08ce4ae6cb8d82af0be1701
	Jiayi Zhang	Aryan Avinash Sawant
	Optimize Visual Effects for Magnetics https://github.com/CSCI-526/main-flip/commit/7f5c0bfde3e3e5bdc1ae1d47808213fbee043bd2b	Implemented Death Spots Metric https://github.com/CSCI-526/main-flip/commit/6b314b4af73849ac7d3cdec0bb51d5319db4c45
Week 13	Xiaoyu Ke	Sherry Wang
	Open area bugs fixed https://github.com/CSCI-526/main-flip/commit/602e5166be4d0859829259db86a14e862ffd29e4	Improved and plotted the graph for metric 4 https://github.com/CSCI-526/main-flip/commit/1fb1c566ec260c3fa450d5e829d513026f9fa1e6
	Xinlu Xu	Yang Li
	Add power-up player reminder and timer panel https://github.com/CSCI-526/main-flip/commit/e58d3ba918546cf1256e3069ac3a9b248b9a62b1	Add the power-up to control hazard https://github.com/CSCI-526/main-flip/commit/e9d265b4af6f442a8dd2419ab08c7f2b04696a78
	Jiayi Zhang	Aryan Avinash Sawant

	Implement key binding modification function https://github.com/CSCI-526/main-flip/commit/2a7343242de537f23e3dc74c62746e2a8ea0c18e	Added Sounds in Game https://github.com/CSCI-526/main-flip/commit/043c4ceb403b071b896d0edc39bd3d8aff7ecaafc
Week 14	Xiaoyu Ke	Sherry Wang
	New Art design and sound design testing https://github.com/CSCI-526/main-flip/commit/49b5808807007850f25a26e8089afa5d72232556	Added the ideal operation value for level 3 https://github.com/CSCI-526/main-flip/commit/29473d43e488062d68836e09629d92b4d0f74553
	Xinlu Xu	Yang Li
	New art design on menu and level page https://github.com/CSCI-526/main-flip/commit/4cbbc2ae92143e46987fb967e4bae06ebf5caf2d	Fixed Autogate bugs in Level 1 and Level 3 https://github.com/CSCI-526/main-flip/commit/e9d265b4af6f442a8dd2419ab08c7f2b04696a78
	Jiayi Zhang	Aryan Avinash Sawant
	Let images and texts sync with key binding https://github.com/CSCI-526/main-flip/commit/df1865a472f5b50a5006ad9f592c6917221cf230	Updated Death Spot Metric https://github.com/CSCI-526/main-flip/commit/362a53466aecec8ff25ed1d798b8e22a7f83534c
Week 15	Xiaoyu Ke	Sherry Wang
	New Art design testing and apply https://github.com/CSCI-526/main-flip/commit/349721859991400bc07cdd3ae5c786077f4543dc	Fixed previous bugs and plotted graphs for metrics https://github.com/CSCI-526/main-flip/commit/97b8d744e6d4e1373d6cae7e3224a42dcb3c523e
	Xinlu Xu	Yang Li
	Adjust background size and plot graphs for metrics https://github.com/CSCI-526/main-flip/commit/da223597704960ff8d3d5622b2082888659a292f	Code Organized & Art implemented https://github.com/CSCI-526/main-flip/commit/5752279aaf6c57181db4fa08ef3412eac79d59a5
	Jiayi Zhang	Aryan Avinash Sawant

	Add introduction UI at gravity tutorial https://github.com/CSCI-526/main-flip/commit/5df98d9b37f90b0635554d45ccaa8604c39bbe63	Updated Levels with Sound branch https://github.com/CSCI-526/main-flip/commit/74755a72f9ed982d8c38919db0191e5c513d7776
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Review Notes

Notes/feedback that you receive from staff and playtesters, as well as possible improvements you could make based on it. It is important that the stenographer makes note of these each week, but other team members should also keep track of incoming feedback.

[Playtest 1: 10/17/2025]

- Trying out the controls first
- Looking for arrow keys
- Stuck on the last platform
- Not sure how gravity works
- Need to show what red and blue color platform means
- Difficulty in magnetism
- Stuck on last platform
- Confused in the colors
- Figuring out as they play few more times
- In the end they got it

[Playtest 2: 10/30/2025]

- Figuring out controls
- Successful in using both mag and grav controls on first attempt.
- Figuring out some platforms respond to gravity shifts.
- Completed LVL 1 in first attempt.
- Didn't recognize what triggers/collectibles are for.
- Don't know how platforms are moving.
- Sticky keys pop up.
- Stuck between checkpoint and 2nd trigger.
- Figured to press shift (mag) twice to repel and attract quickly.
- Success after 4-5 tries.
- Figured out on the 2nd trigger that the certain platforms respond to trigger collection.
- Level 2 is too difficult.

- Stuck for some time on the last part.
- In the end figured out
- Pop up clue if stuck for too long (suggestion)
- Good game in general (feedback)

[Playtest 3: 11/07/2025]

- 00:05 - Reading the hints presented.
- 00:15 - Successfully completed tutorial.
- 00:40 - Sticky keys popup.
- 01:00 - Polarity change.
- 01:18 - Death caused by the trap.
- 02:45 - Stuck at lvl 1.
- 02:50 - Fig that the player has to be in the centre.
- 03:25 - Completed that stage.
- 05:00 - Stuck on the next stage.
- 06:00 - Mentioned "need to time it properly".
- 06:30 - Went to tutorial.
- 06:50 - Jumped to lvl 3.
- 07:05 - Then back to lvl1.
- 07:45 - STILL STUCK AT LVL 1.
- 08:30 - Slowly figuring it out to time it.
- 09:20 - Done with that stage.
- 09:50 - Last stage of Level 1.
- 10:15 - Completed Level 1.

[Playtest 4: 11/21/2025] Playtesters: Team DOS

- Starting with the tutorial.
- Reading the instruction
- Liked the green/red arrow
- Didn't figure the relaxation time on the polarity and gravity buttons.
- Death in last part lvl 1
- Level 1 next lvl not working
- Respawn not working on lvl 2 clicking the button
- Figured that they can change key bindings which rectifies the sticky key pop up bug

[Playtest 4: 11/21/2025] Playtesters: SKILLLIMITED

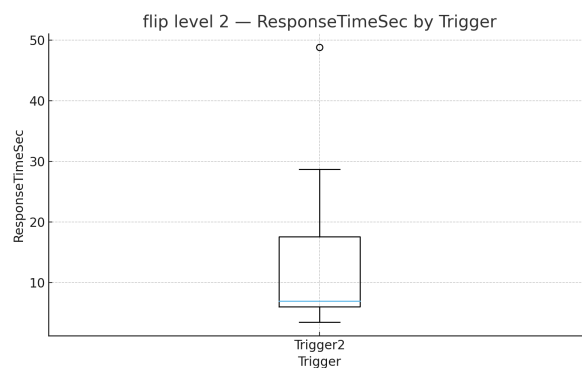
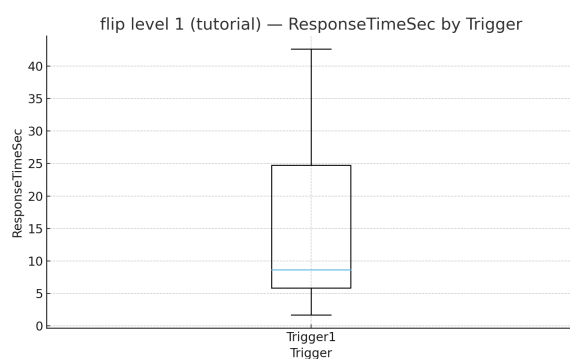
- Feedback: in beta feedback mentioned about changing the position of polarity and gravity buttons on the UI. Which was taken care of in gold play testing.
- increase the rotation speed of the platform in level 1 at the end.
- Level 1 next lvl not working.
- Completed all the three levels.

[Playtest 4: 11/21/2025] Playtesters: QWERTY

- The Respawn button became unresponsive in the middle section of Level 1.
- The Next Level button from Level 1 to Level 2 did not work.
- Players felt uncomfortable using the charge bar.

[<Alpha Milestone> Playtest results]

- Analytics



- The Trigger Response Time for **Trigger 1** in Level 1 met our expectations.
- The Trigger Response Time for **Trigger 2** in Level 2 exceeded our anticipation, so we added a clear visual hint—a **camera zoom-in**—in the updated level design to help players identify the correct path more easily.

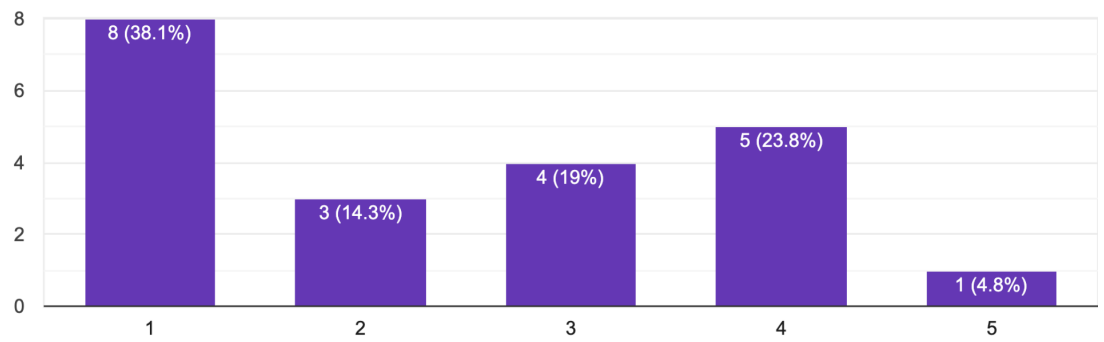
Alpha Survey

- <https://docs.google.com/forms/d/e/1FAIpQLSfiwdGR7i5miFM6jJdSAczkUCfxyL13LDvLSgcxS1q3xSaOKg/viewform?usp=dialog>

1: Easy and 5: Very hard

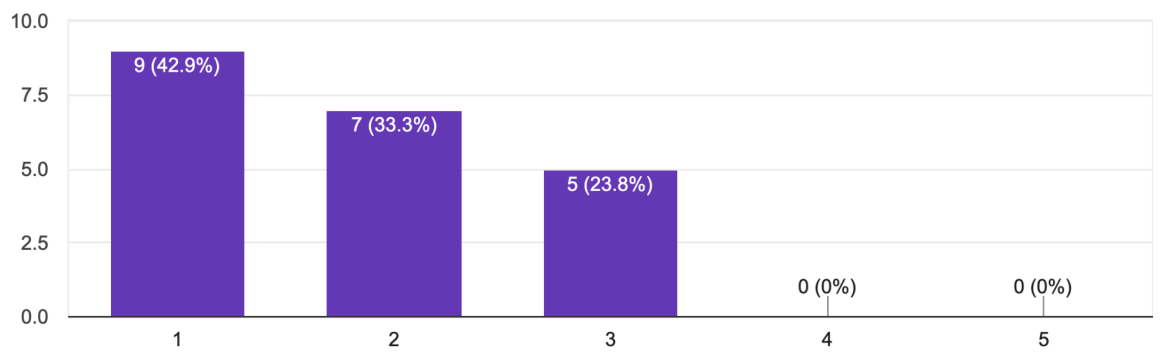
How easy was it for you to learn how to play the game?

21 responses



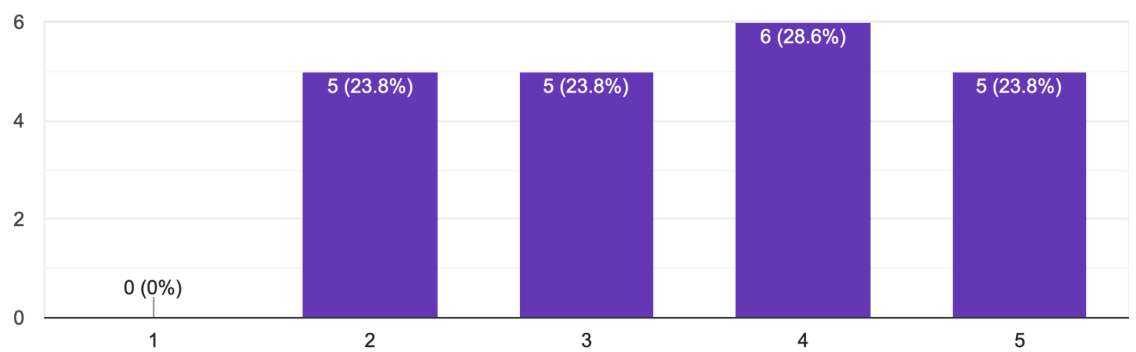
How difficult is the first level?

21 responses



How difficult is the second level?

21 responses



Favorite Parts of the Game (Strengths)

- **Core Mechanics:** The central idea of the game is highly appreciated.
 - **Novelty:** Players liked the "**Loved the idea,**" "**novel way of control,**" and "**creativity.**"
 - **Polarity/Gravity:** The specific mechanics of "**switch between north pole and south pole**" and the "**mechanisms of gravity and magnetic force and their combined effects**" are key draws.
- **Aesthetic & Feel:**
 - **Satisfaction:** Players enjoy the "**feel really good**" moment when the Gravity and Magnetism switches are successful.
 - **Visuals:** Liked the "**clean UI aesthetics**" and the "**design of the gravity and the magnetic force.**"
 - **View:** Appreciated the "**Auto view centering upon entering different stages.**"
- **Specific Level Design:**
 - Players enjoyed the challenges, specifically mentioning the "**moving magnets that make players think**" and the "**excitement of avoiding the ball contacting the spikes.**"
 - General praise for the "**Level Design.**"

Least Favorite Parts of the Game (Weaknesses/Friction)

- **Technical/Accessibility Issues (CRITICAL):**
 - The **Sticky Keys pop-up** when playing is a significant and noted pain point. (This directly relates to your previous question).
 - **Hitbox Issues:** Player mentioned "**Some hitbox**" problems.
- **Difficulty & Pacing:**
 - **Level 2 Spike:** The difficulty jump is too steep, with multiple comments citing the "**difficulty of the second level**" and that the difficulty "**seems to be increased a little bit too much in level 2.**"
 - **Frustration:** Comments like "**think,**" "**Little confusing,**" and "**Felt like I was constantly fighting to move forward**" indicate friction.
- **Mechanics & Flow:**
 - **Momentum:** The inability to "**carry momentum through sections**" is frustrating and affects the flow.
 - **Switches/Puzzles:** The switches in Level 2 are "**not very well explained**" and players have "**no idea what they did otherwise.**"
- **Game Flow:**
 - Lack of content/progression beyond the test levels: "**No next level after 2.**"

Suggested Improvements (Action Items)

- **Tutorial & Onboarding (HIGH PRIORITY):**
 - The overwhelming request is for better instruction: **"A small tutorial," "have a guided tutorial," and "give a little bit of instructions for beginners."**
 - **Switch Clarity:** Switches need to be **"more clearly explained and also have a way to show what they do."**
- **Quality of Life & Visual Feedback:**
 - **Gravity Feedback:** Need a **"better visual to see if the gravity is up or down instead of looking at the bottom left."**
 - **Spawning Issue:** Investigate the bug where a player **"died and spawned at a checkpoint even though I didn't get there."**
 - **Hitbox/Physics:** **"Improve hitbox and physics."**
- **Difficulty Tuning (Pacing):**
 - **Confidence Building:** Needs **"a bit more room for screwing up so players can build confidence."**
 - **Intermediate Level:** Suggestion to insert **"a level with difficulty that is medium between level 1 and 2."** (This aligns with the difficulty spike criticism).
- **Future Features:**
 - **Competitive:** Add a **"timer and a leader board"** for speedrunners.
 - **Educational:** **"Include some physical knowledge related to magnetic and gravitational forces."**
- **Mobile Specific:**
 - **Controls:** Need **"Larger buttons for mobile players!"**

Hypothesized Issues from Feedback (Gold):

Hypothesized Issue #1

Sticky Keys Pop-up while playing.

Description of the issue

The primary issue was the frequent and disruptive appearance of the Windows operating system's "Sticky Keys" pop-up during intense gameplay moments. This pop-up, which is triggered by pressing the Shift key five times in quick succession, halted the game, interrupted player input, and pulled the player's focus away from the screen, leading to frustration and perceived unfairness, especially in timing-critical situations.

Significance

This issue was highly significant because it was a **critical friction point** that severely impacted the core gameplay loop and player retention. It directly violated the principle of flow by repeatedly breaking immersion. If left unaddressed, it would lead to negative player reviews and a high rate of early play session abandonment, as players would perceive the game as "broken" or "unresponsive" rather than recognizing it as an OS-level feature.

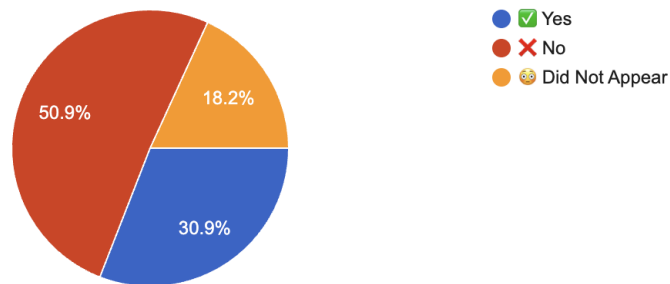
Feedback/Data points

- **Significant Exposure Rate:** Over **30%** (specifically 30.9%) of all respondents reported that the Sticky Keys pop-up **did appear** while they were playing. This high rate indicates that the issue was not isolated to a small subset of users but was a widespread problem affecting nearly a third of your player base.
- **High Impact on Gameplay:** Of the players who encountered the pop-up, a significant majority (**56.4%**) confirmed that it **negatively affected their gameplay experience**. This confirms the issue was not just a minor annoyance but a disruptive factor that broke player flow and potentially led to in-game failure.
- **Validation of Fix:** The combination of a high occurrence rate and high impact rating provides **strong statistical validation** that the decision to implement a **Key Binding** feature was a necessary and high-priority fix to improve core player experience and retention.
- **In Player Action Switch Rate Metric,** we noticed that in the levels where the switch rate between the controls is high, there was a higher chance of sticky keys pop-up as the player has to press the keys quickly. Which is something we also noticed during the in-class playtesting.

Did the Sticky Keys popup appear while playing?

[Copy chart](#)

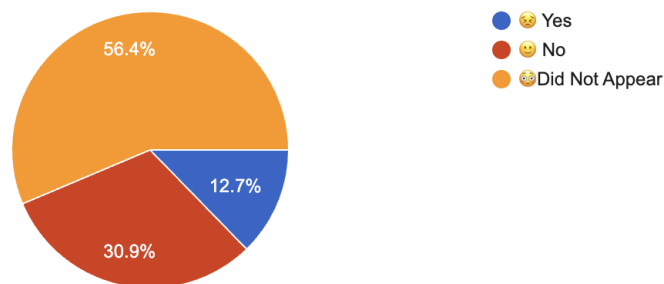
55 responses



If "Yes" to the previous question, did it affect your gameplay experience?

[Copy chart](#)

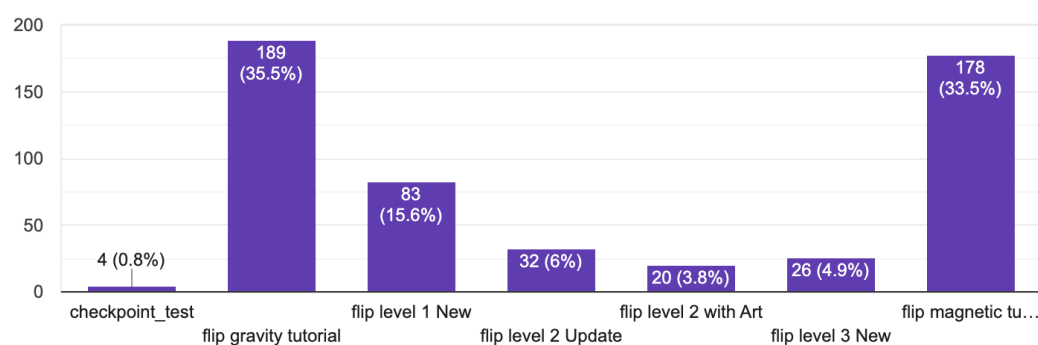
55 responses



Scene

[Copy chart](#)

532 responses



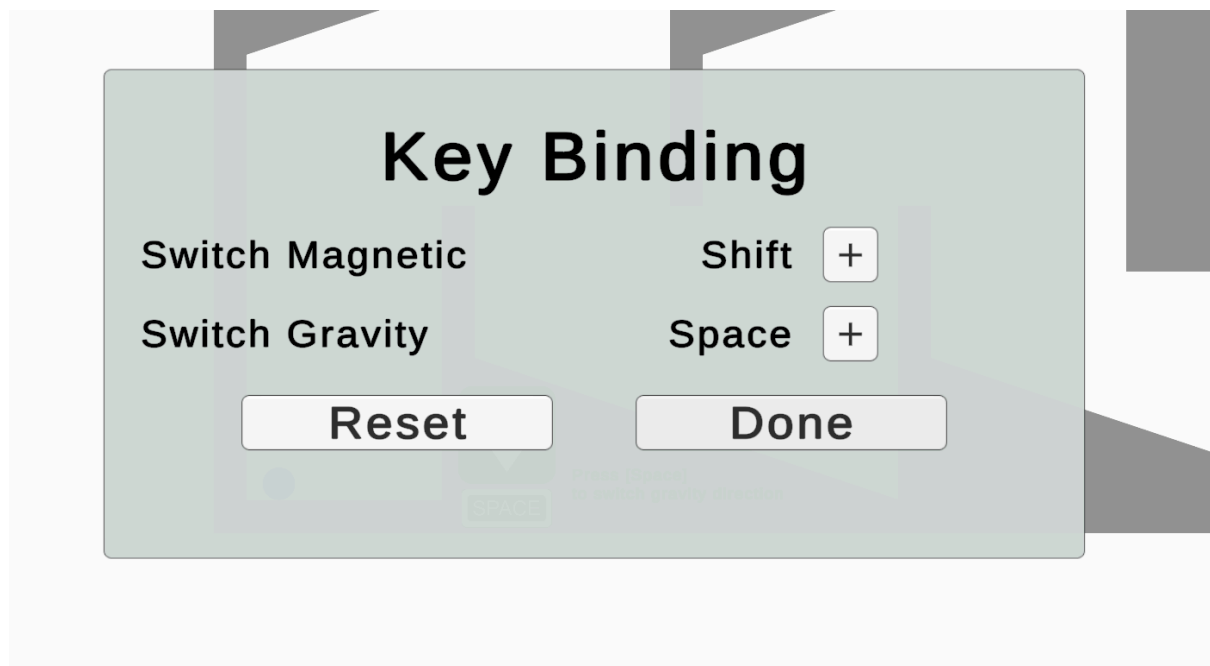
Potential Fixes

- *Change Default Binding*
- *Implement Custom Key Binding*

Implemented Solution

Implement Custom Key Binding (Flexible/Player-Controlled): Introduce an in-game

settings option to allow players to re-map any action to a key of their choice.



Hypothesized Issue #2

*"Not easy for beginners". The core issue is the **insufficient tutorialization** of key game mechanics and controls, particularly at the start of the first major level (Level 1).*

Description of the issue

The core issue is the insufficient tutorialization of key game mechanics and controls, particularly at the start of the first major level (Level 1). This lack of proper guidance leaves new players ill-equipped to handle the initial obstacles, resulting in a high rate of early-level failure and frustration. This problem is succinctly captured by player feedback stating: "Not easy for beginner."

Significance

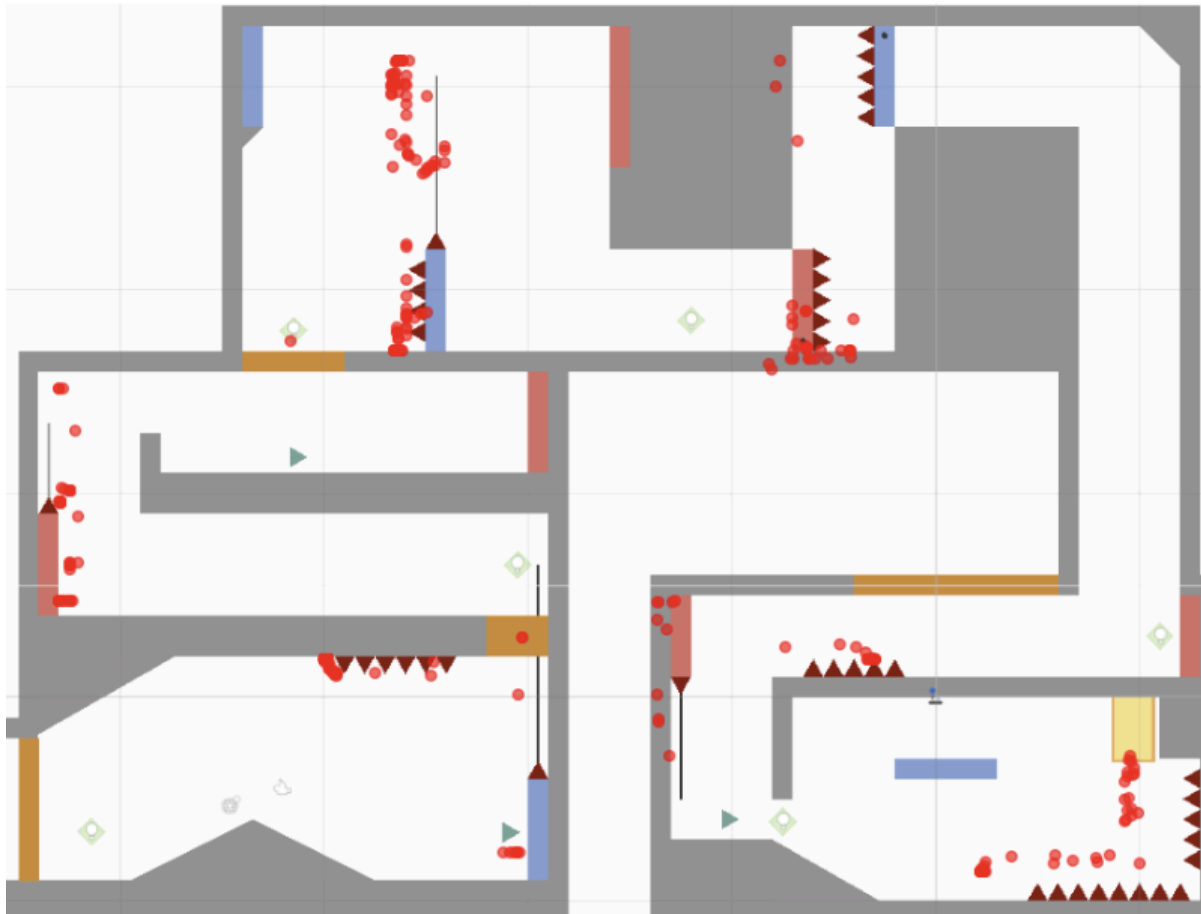
*This issue is highly significant as it represents a **failure in player onboarding**. If players die repeatedly at the very beginning of the first level, they are likely to perceive the game as unfair or confusing, leading to high abandonment rates and failure to progress. Effective tutorialization is crucial for establishing the game's language and ensuring players understand the tools necessary for success.*

Feedback/Data points

- **Analytics Data (Graph Citation):** The **Death Spots Metrics** clearly show that **"many players were dying at a very early stage of Level 1."** This quantitative data indicates that players are struggling with the very first challenges they face, suggesting they either don't know the required solution or lack familiarity with

the necessary controls.

- **Survey Feedback (Feedback Citation):** The qualitative feedback confirms the cause of the deaths. **"Many players mentioned that there should be more tutorialization which will help the player understand the mechanics."** This directly links the high death rate to a knowledge gap, rather than an issue with the level design itself.



Tutorial is quite short.

hard to follow at first

Its very challenging

There's a learning curve to understanding the polarities and gravity, as it should be

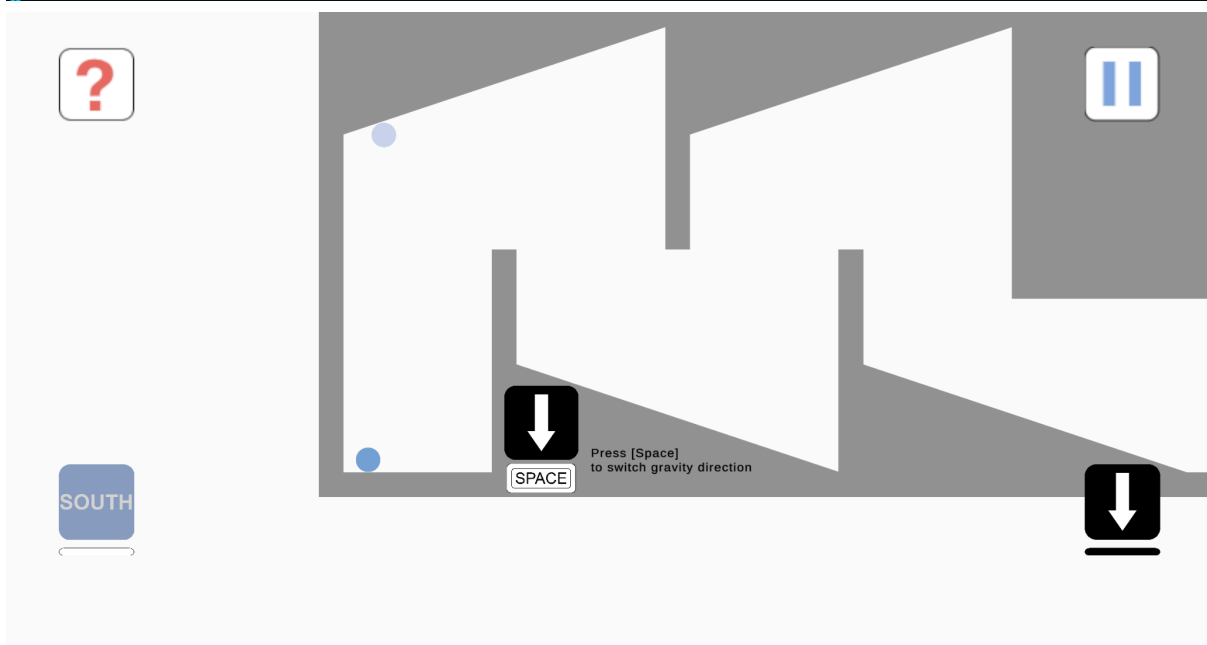
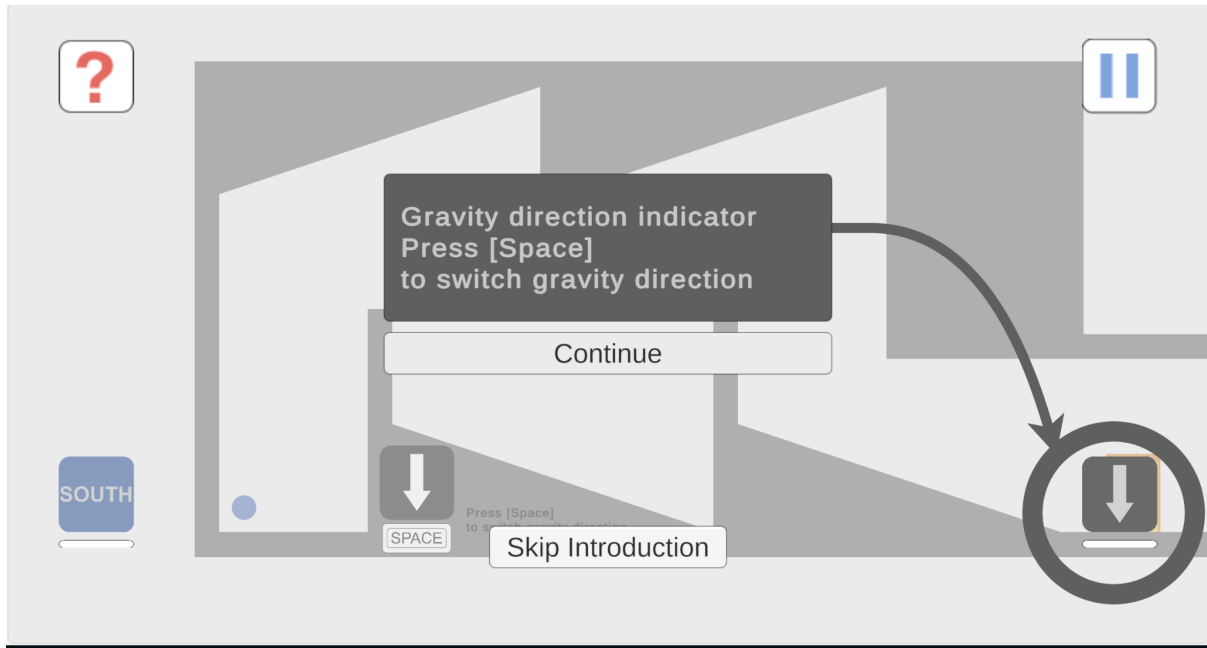
Potential Fixes

- *Reduce Early Difficulty (Design Rework)*
- *Integrated Contextual Tooltips (In-Level Guidance)*
- *Dedicated Pre-Level Tutorial*

Implemented Solution

- *Solution A: Reduce Early Difficulty (Design Rework): Simplify the initial obstacles*

- in Level 1 to be less demanding, compensating for the lack of knowledge.
- **Solution B: Integrated Contextual Tooltips (In-Level Guidance):** Add small, non-intrusive on-screen text prompts that appear only when a new mechanic is introduced in Level 1.
 - **Solution C: Dedicated Pre-Level Tutorial (Mandatory Education):** For this we created separate tutorials, one for gravity shift and another for polarity switch. We also implemented a ghost player which directs the player to a path which the player can follow to complete the tutorial.



Hypothesized Issue #3

Player Stuck States and Lack of Directional Clarity.

Description of the issue

The primary issue was the prevalence of **player stuck states** within the game environment, where players would become physically trapped (e.g., clipping into geometry or falling into non-lethal, non-returnable areas) or functionally stuck due to a **lack of navigational clarity** (i.e., not knowing where to go next). This forced players to quit and restart the level, breaking immersion and causing significant frustration.

Significance

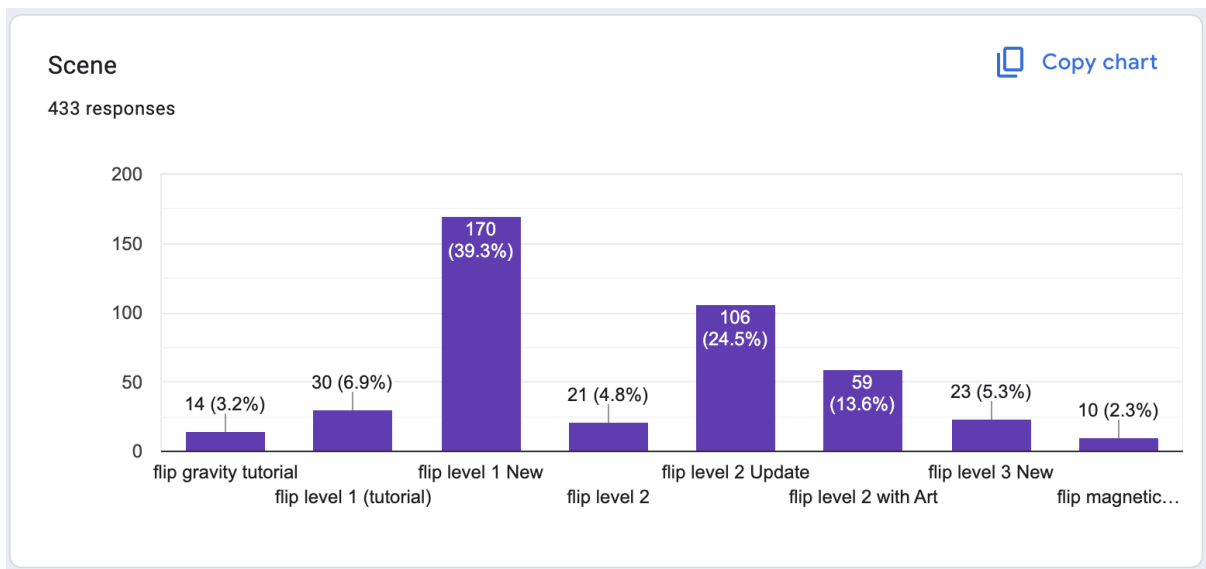
This issue is critical because it leads to involuntary session termination, which is highly detrimental to player experience. Getting stuck due to a technical flaw (like clipping) or confusing design (lack of direction) makes the player feel lost and helpless. The resulting restarts waste player time and directly contribute to negative sentiment and eventual abandonment of the game.

Feedback/Data points

Analytics Data (Graph Citation): The **Trigger Response Time Metric** showed abnormally **long response times** between Trigger N and the activation of the corresponding mechanism in specific non-puzzle areas. Since the purpose of this metric is to check if triggers are placed efficiently and are intuitive, the prolonged response times specifically in the level 1 indicated two things:

1. The player **could not locate** the next trigger (navigational confusion/lost state).
2. The player was **physically trying to escape** a stuck position before resuming progress.

Survey Feedback (Feedback Citation): The problem was confirmed by direct accounts: **"There were a couple of feedbacks that said that they were stuck at some point and couldn't move."** This qualitative feedback links the long response times to both physical trapping and functional confusion.



You can get stuck in someplaces and changing gravity or magnetism does not help you move

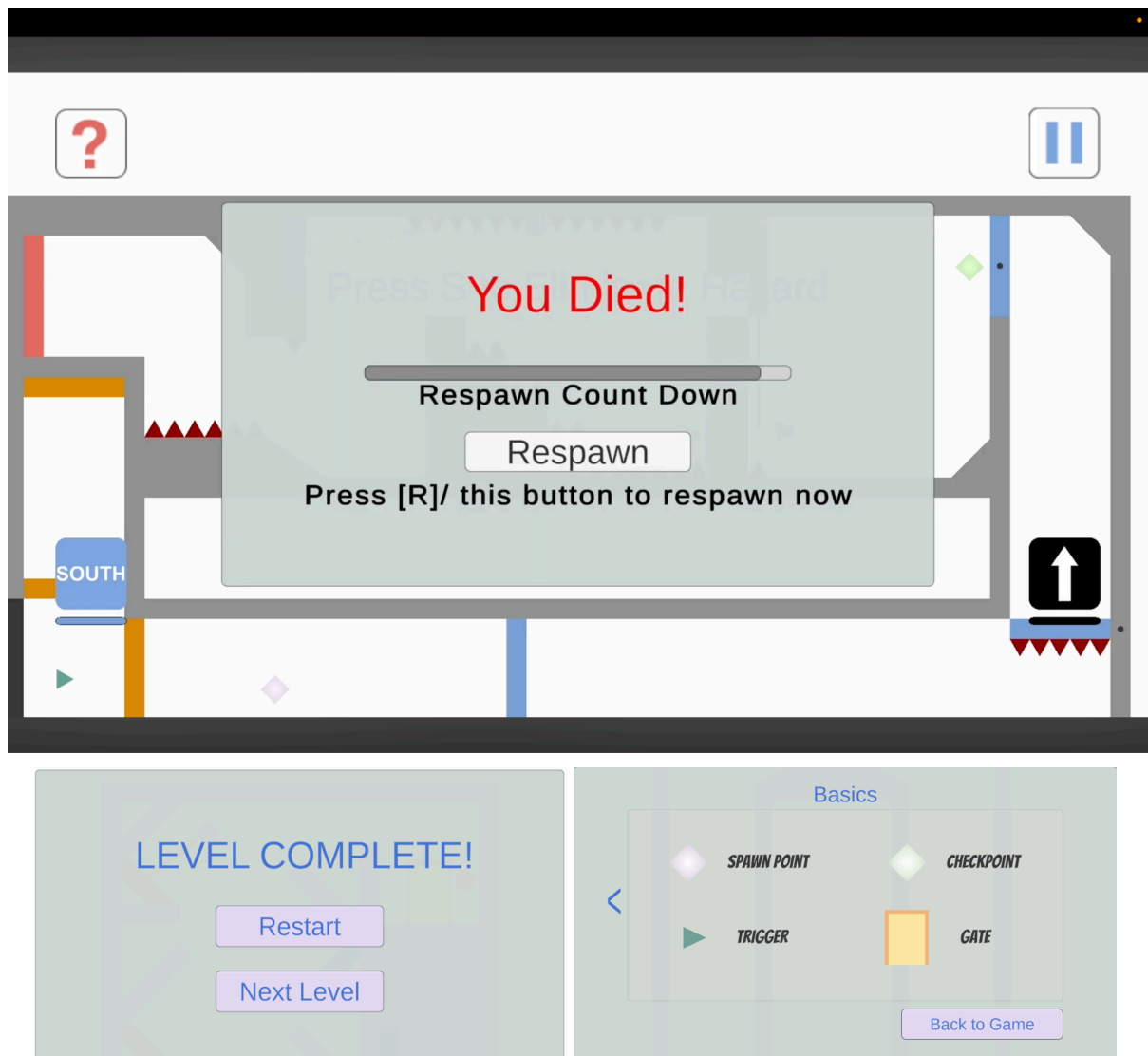
unintentional movement causing a reset(level 2 middle section, shifting gravity when i'm attached to a magnet cause me to oscillate a bit and touched the spikes)

Potential Fixes

- *Add Quick Respawn Button*
- *Redesign Problematic Geometry*
- *Dynamic FOV Camera Focus.*

Implemented Solution

- *Dynamic FOV Camera Focus: Implementing a dynamic FOV camera that focuses on the player's immediate next path is the most effective way to address the navigational confusion suggested by the prolonged Trigger Response Times. This subtle cinematic guidance naturally directs the player's eye, showing them where the next trigger or mechanism is located without needing intrusive UI arrows.*
- *Immediate Relief (Stuck State): Adding a Respawn Button provides a reliable emergency exit for technical stuck states, allowing the player to reset from a bad position quickly.*
- *Accessibility: The addition of an Options UI showing controls provides a quick reference, further empowering players to utilize necessary mechanics to overcome obstacles.*



Hypothesized Issue #4

Difficulty Spike and High Death Concentration in Level 3

Description of the issue

The issue was a localized, high concentration of player deaths at a specific, difficult segment in Level 3. This particular section created a significant difficulty spike, causing players to fail repeatedly and resulting in acute frustration and potential session abandonment before they could progress past that point.

Significance

This issue was highly significant because it represented a choke point in the player progression funnel. While difficulty is intended, a sudden and repeated failure at one spot suggests an unbalanced challenge that may push players past their frustration tolerance, leading to higher dropout rates specifically at the threshold of Level 3.

Feedback/Data points

The problem was identified through clear death mapping and inferred player frustration:

- **Analytics Data (Graph Citation):** The Death Spots Metrics clearly showed an abnormal and severe cluster of death markers concentrated at one particular section of Level 3. This quantitative data precisely identified the location and magnitude of the difficulty spike.
- **Survey Feedback (Feedback Citation):** While specific feedback for this spot might be scarce, the general sentiment regarding difficulty (like "Not easy for beginner" or requests for reduced difficulty/checkpoints, as seen in Issue 2) supports the idea that players required intervention for extreme difficulty spikes. The high death count itself serves as the most critical player feedback, indicating the challenge was tuned too harshly.



The difficulty of the second level.

The second level felt too hard

Second level tooooooo difficult 😞

not easy for beginners

Potential Fixes

Reduce Spike Count/Rework Spikes

Add Checkpoint Immediately Before Spikes

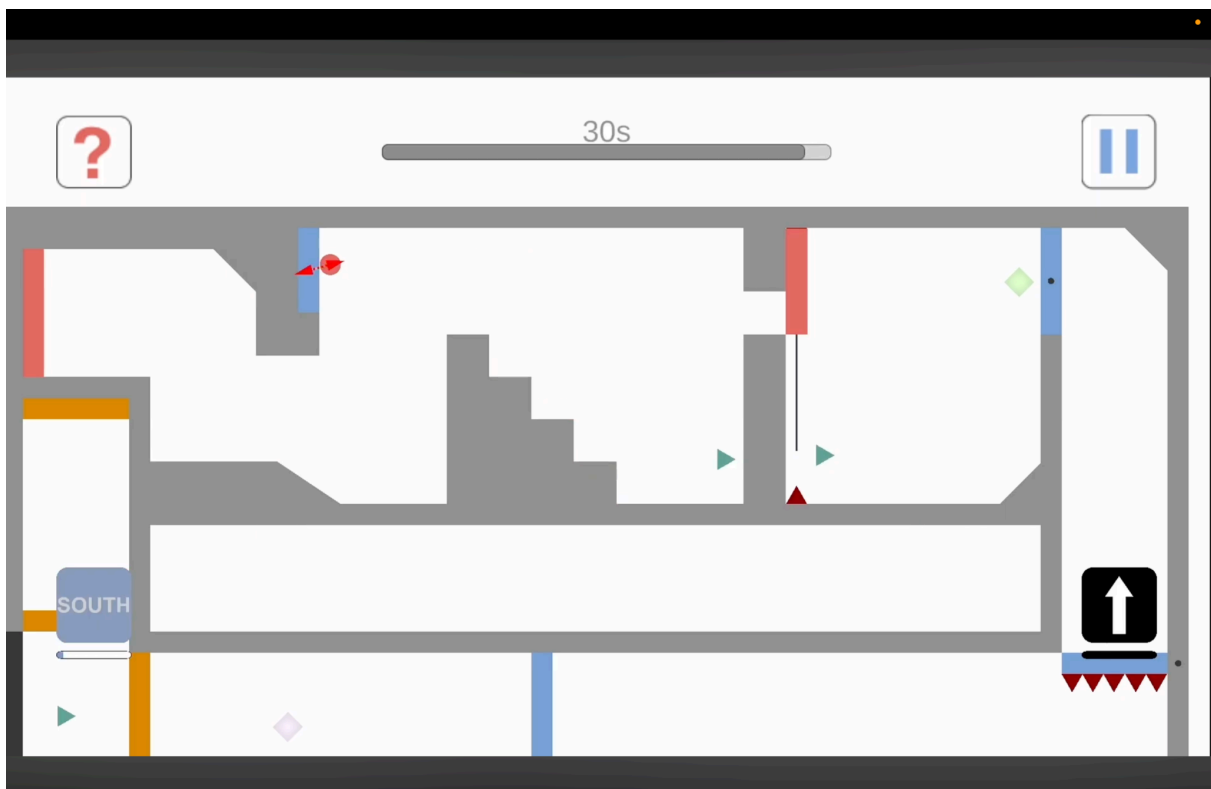
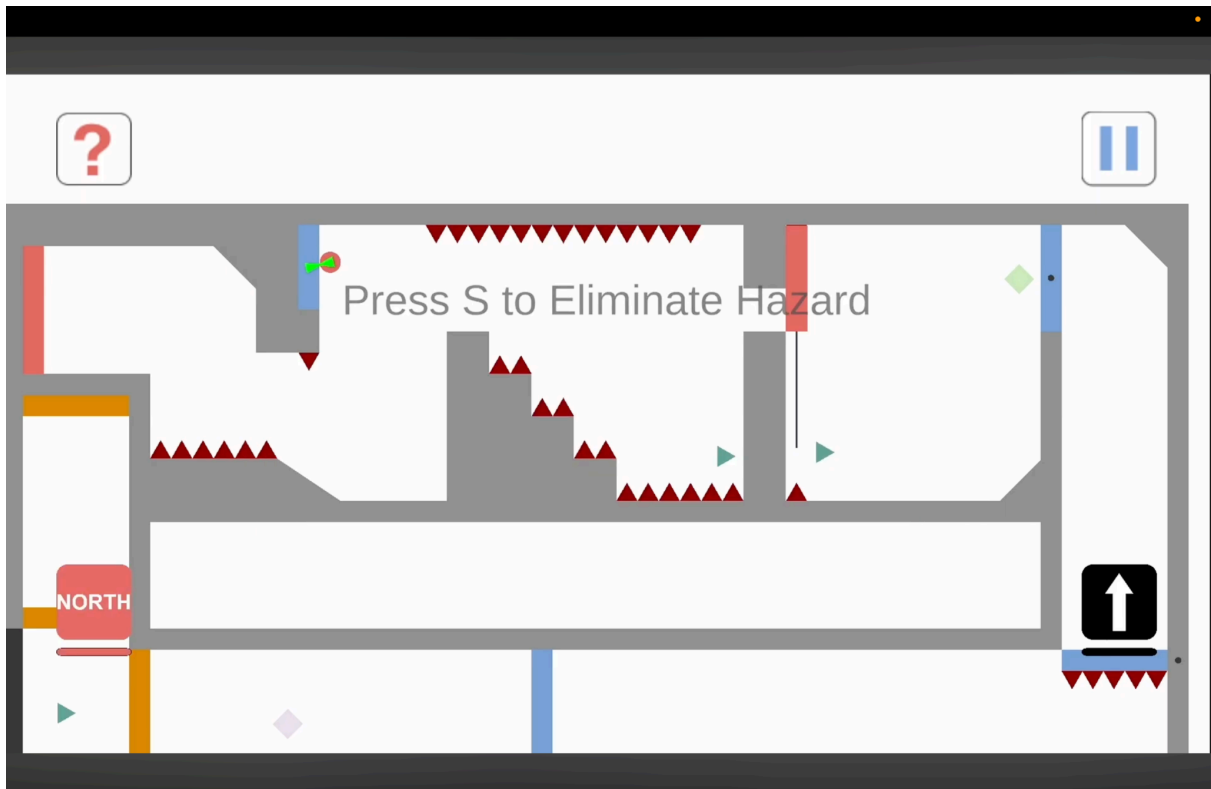
Implement Adaptive Assistance Mode

Implemented Solution

Implementing an Adaptive "Safe Mode."

Why this solution was chosen:

- *Maintains Challenge Integrity: Unlike permanent design changes (Solution A), the Safe Mode preserves the intended challenge for players who can clear it on their first few attempts. It only provides assistance to those who are struggling.*
- *Directly Addresses Frustration: The trigger—dying 5 times—ensures the assistance is only offered after the player has experienced acute frustration, providing a graceful mechanism for them to advance without quitting.*
- *Mechanism: When the player dies 5 times, they are prompted to press S to activate the mode, which makes all spikes disappear for 30 seconds, guaranteeing a window for successful progression.*



External References

Name	Link	Usage
<i>Python Libraries: gspread & matplotlib</i>	https://www.python.org/	<i>Used for scaling and plotting</i>
<i>Google Sheets API</i>	https://developers.google.com/sheets/api	<i>Used to read/write data from Google Sheets.</i>
<i>Google Drive API</i>	https://developers.google.com/drive	<i>Used to access and manage files in Google Drive.</i>
<i>Google Apps Script</i>	https://developers.google.com/apps-script	<i>Used to automate tasks in Google Sheets and integrate with APIs.</i>
<i>Background Music</i>	https://soundimage.org/chiptunes-4/	<i>Background Music.</i>
<i>Zaplast.com</i>	https://www.zapsplat.com/sound-effect-category/button-clicks/	<i>Sounds in game.</i>
<i>kenney.nl</i>	https://kenney.nl/assets/pico-8-platformer	<i>Spikes in Level 2</i>
<i>kenney.nl</i>	https://kenney.nl/assets/pixel-platformer-industrial-expansion	<i>Platforms and Background picture in Level 2</i>
<i>itch.io</i>	https://omniclause.itch.io/16x16-floppy-disk-spinning-animation	<i>Checkpoints</i>
<i>itch.io</i>	https://hazushadc.itch.io/pixel-door-assets	<i>Gate</i>

Appendix